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Paper No. 49
Date: May 17,2023

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SYNTHEGO CORPORATION,
Petitioner,

v.

AGILENT TECHNOLOGIES, INC.,
Patent Owner.

IPR2022-00403
Patent 10,900,034 B2

Before ROBERT A. POLLOCK, DAVID COTTA, and
MICHAEL A. VALEK, *Administrative Patent Judges.*

VALEK, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

Granting Joint Motion to Seal
Granting Patent Owner's Motion to Seal Portions of Sur-reply Brief
37 C.F.R. §§ 42.14, 42.54

I. INTRODUCTION

Synthego Corporation ("Petitioner") filed a Petition (Paper 1, "Pet."), seeking *inter partes* review of claims 1–33 of U.S. Patent No. 10,900,034 B2 (Ex. 1001, "the '034 patent"). We instituted trial on all of the grounds in the Petition. Paper 11, 31.

Following institution, Agilent Technologies, Inc. ("Patent Owner") filed a Response (Paper 18, "Resp."), Petitioner filed a Reply (Paper 30, "Reply"), and Patent Owner filed a Sur-reply (Paper 33, "Sur-reply"). We held a hearing on March 1, 2021, and a transcript is of record. Paper 48 ("Tr.").

In addition, the parties have jointly moved to seal Exhibits 1053–1058 and portions of the Reply (Paper 29) and Patent Owner has moved to seal portions of the Sur-reply (Paper 34).

After considering the parties' arguments and evidence, we find that Petitioner has shown by a preponderance of the evidence that the challenged claims of the '034 patent are unpatentable. *See* 35 U.S.C. § 316(e). We also grant both motions to seal. Our reasoning is explained below.

II. BACKGROUND

A. Real Parties in Interest

Petitioner and Patent Owner identify themselves as the only real parties in interest. Pet. 15; Paper 4, 2.

B. The '034 Patent

The '034 patent issued on January 26, 2021, and claims priority to a utility application filed on December 3, 2015, as well as a series of provisional applications the earliest of which was filed on December 3, 2014. Ex. 1001, codes (60) (63).

The ’034 patent relates to “modified guide RNAs and their use in clustered, regularly interspaced, short palindromic repeats (CRISPR)/ CRISPR-associated (Cas) systems.” Ex. 1001, Abstr. The Specification explains that “[i]n the native prokaryotic system” from which CRISPR technology is derived “the guide RNA (‘gRNA’) comprises two short, non-coding RNA species referred to as CRISPR RNA (‘crRNA’) and trans-acting RNA (‘tracrRNA’).” *Id.* at 1:41–44. The native CRISPR-Cas system may also be engineered to use a single guide RNA (sgRNA) that combines the crRNA and tracrRNA into a single molecule. *Id.* at 1:57–59. The guide RNA forms a complex with a Cas nuclease that is able to bind to a target DNA site adjacent a protospacer adjacent motif (“PAM”) sequence and cleave the target DNA at that specific site. *Id.* at 1:44–49, 2:21–34; *see also* Ex. 1003 ¶¶ 44–48; Ex. 2003 ¶¶ 49–54 (declarant testimony from both parties offering similar technical background on guide RNA and its function in CRISPR-Cas systems).

According to the Specification, “there is a need for providing gRNA, including sgRNA, having increased resistance to nucleolytic degradation, increased binding affinity for the target polynucleotide, and/or reduced off-target effects while, nonetheless, having gRNA functionality.” Ex. 1001, 2:4–8. The Specification states that Patent Owner’s “invention is based, at least in part, on an unexpected discovery that certain chemical modifications to gRNA are tolerated by the CRISPR-Cas system.” *Id.* at 3:51–53. These modifications are “believed to increase the stability of the gRNA, to alter the thermostability of a gRNA hybridization interaction, and/or to decrease the off-target effects of Cas:gRNA complexation" and "do not substantially compromise the efficacy of Cas:gRNA binding to, nicking of, and/or cleavage of the target polynucleotide." *Id.* at 3:50–59.

C. Challenged Claims

The Petition challenges claims 1–33. Of these, claims 1 and 19 are independent. Claims 1 and 19 are illustrative and read as follows:

> 1. A synthetic CRISPR guide RNA comprising:
> (a) a crRNA segment comprising (i) a guide sequence capable of hybridizing to a target sequence in a polynucleotide, (ii) a stem sequence; and
> (b) a tracrRNA segment comprising a nucleotide sequence that is partially or completely complementary to the stem sequence,
> wherein the synthetic guide RNA has gRNA functionality comprising associating with a Cas protein and targeting the gRNA:Cas protein complex to the target sequence, and comprises one or more modifications in the guide sequence wherein the one or more modifications comprises a 2'-O-methyl.
>
> 19. A synthetic CRISPR crRNA molecule comprising
> a guide sequence capable of hybridizing to a target sequence in a polynucleotide, wherein the synthetic crRNA molecule comprises one or more modifications in the guide sequence;
> wherein the synthetic crRNA molecule has gRNA functionality comprising associating with a Cas protein and targeting the gRNA:Cas protein complex to the target sequence; and wherein the one or more modifications comprises a 2'-O-methyl.

Ex. 1001, 257:34–47, 259:1–9.

D. Asserted Grounds of Unpatentability

Petitioner asserts the following grounds of unpatentability:

Claim(s) Challenged	35 U.S.C. §[1]	Reference(s)/Basis
1–5, 8–21, 24–33	102	Pioneer Hi-Bred[2]
5, 8–13, 18, 20, 21, 24–28, 32	103	Pioneer Hi-Bred and Krützfeldt,[3] Deleavey,[4] Soutschek,[5] or Yoo[6]
6, 7, 22, 23	103	Pioneer Hi-Bred and Threlfall[7] or Deleavey
3, 4	103	Pioneer Hi-Bred and Knowledge of Person of Ordinary Skill in the Art ("POSA")
5, 8–13, 20, 21, 24–28	103	Pioneer Hi-Bred and Knowledge of POSA

[1] The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) ("AIA"), included revisions to 35 U.S.C. §§ 102 and 103 that became effective prior to the filing of the application that led to the '001 patent. Therefore, we apply the AIA versions of 35 U.S.C. §§ 102 and 103.

[2] WO 2015/026885 A1, published February 26, 2015 (Ex. 1006) ("Pioneer Hi-Bred").

[3] Jan Krützfeldt et. al, "Specificity, Duplex Degradation and Subcellular Localization of Antagomirs," 35 Nucleic Acids Research 2885–2892 (2007) (Ex. 1009) ("Krützfeldt").

[4] Glen F. Deleavey et. al., "Designing Chemically Modified Oligonucleotides for Targeted Gene Silencing," 19 Chem. & Bio. Review 937–954 (2012) (Ex. 1007) ("Deleavey").

[5] Jürgen Soutschek et. al., "Therapeutic Silencing of an Endogenous Gene by Systemic Administration of Modified siRNAs," 432 Nature 173–178 (2004) (Ex. 1012) ("Soutschek").

[6] Byong Hoon Yoo et al., "2′-*O*-methyl-modified Phosphorothioate Antisense Oligonucleotides Have Reduced Non-specific Effects *In Vitro*," 32 Nucleic Acids Research 2008–2016 (2004) (Ex. 1011) ("Yoo").

[7] Richard N. Threlfall et al., "Synthesis and Biological Activity of Phosphonoacetate- and Thiophosphonoacetate-modified 2′-*O*-methyl Oligoribonucleotides," 10 Org. Biomol. Chem., 746–754 (2012) (Ex. 1010) ("Threlfall").

Claim(s) Challenged	35 U.S.C. §[1]	Reference(s)/Basis
14, 29	103	Pioneer Hi-Bred and Knowledge of POSA

In support of these grounds, Petitioner relies on declarations from Henry Morrice Furneaux submitted with the Petition (Ex. 1003) and Reply (Ex. 1059). Patent Owner relies on declarations from William S. Marshall submitted with its Preliminary Response (Ex. 2003) and Response (Ex. 2025). Patent Owner also relies on a declaration from one of the inventors named on the ’034 patent, Jeffrey R. Sampson (Ex. 2029).

III. ANALYSIS OF THE ASSERTED GROUNDS

A. Legal Standards

“In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016). This burden of persuasion never shifts to patent owner. *See Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015).

i. Anticipation

To establish anticipation, each limitation in a claim must be found in a single prior art reference, arranged as recited in the claim. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1369 (Fed. Cir. 2008). Although the elements must be arranged or combined in the same way as in the claim, “the reference need not satisfy an *ipsissimis verbis* test.” *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009).

Further, to be anticipating, a prior art reference must be enabling. *In re Morsa*, 803 F.3d 1374, 1377 (Fed. Cir. 2015). "Enablement of prior art requires that the reference teach a skilled artisan—at the time of filing—to make or carry out what it discloses in relation to the claimed invention without undue experimentation." *Id.* (citing *In re Antor Media Corp.*, 689 F.3d 1282, 1289–90 (Fed. Cir. 2012)). Prior art disclosures are presumed enabling. *In re Antor Media Corp.*, 689 F.3d 1282, 1287–88 (Fed. Cir. 2012); *Apple Inc. v. Corephotonics, Ltd.*, 861 Fed. Appx. 443, 450 (Fed. Cir. 2021) ("[R]egardless of the forum, prior art patents and publications enjoy a presumption of enablement, and the patentee/applicant has the burden to prove nonenablement for such prior art.").

ii. Obviousness

A claim is unpatentable for obviousness if, to one of ordinary skill in the pertinent art, "the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains." 35 U.S.C. § 103; *see also KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) when in evidence, objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

Subsumed within the *Graham* factors is the requirement that the skilled artisan would have had a reasonable expectation of success in combining the prior art references to achieve the claimed invention. *Pfizer, Inc. v. Apotex, Inc.*, 480 F.3d 1348, 1361 (Fed. Cir. 2007). "Obviousness does not require absolute predictability of success . . . all that is required is a reasonable expectation of success." *In re O'Farrell*, 853 F.2d 894, 903–4 (Fed. Cir. 1988). Moreover, "[t]he combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results." *KSR*, 550 U.S. at 416.

On the other hand, a patent claim "is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art." *KSR*, 550 U.S. at 418. An obviousness determination requires finding "both 'that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.'" *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation omitted).

B. Level of Ordinary Skill in the Art

Relying on the testimony of its declarant, Dr. Furneaux, Petitioner contends that a POSA "as of December 3, 2014 (the earliest possible priority date of the '034 Patent) would have had a Ph.D. in molecular biology, biochemistry, or a related discipline." Pet. 11 (citing Ex. 1003 ¶ 59). Petitioner further asserts that "[a] POSA would have understood the chemical structure of nucleic acids, such as DNA and RNA, and would have understood the role of such nucleic acids in cellular biology" and been aware of various prior art "methods to chemically modify RNA, including gRNAs for use in gene regulation." *Id.* at 11–12 (citing Ex. 1003 ¶¶ 60–69). Petitioner further asserts that a POSA would "have known about uses of chemically modified gRNA in CRISPR-Cas applications," "about making libraries of such modified gRNAs," and "attaching fluorophores at the 5'-ends . . . for purposes of tracking them." *Id.* at 12–13 (citing Ex. 1003 ¶¶ 70–80).

Patent Owner agrees that a POSA would have this educational level. Resp. 23. Patent Owner "also agrees that a POSA would have knowledge of prior art RNA based gene regulating technologies, but disagrees that the teachings of those technologies are relevant in the manner that Dr. Furneaux [and Petitioner] attempt[] to apply them" in the Petition's grounds. *Id.* at 23–24.

We find the parties' agreed understanding of the level of ordinary skill in the art to be supported by the record and apply that description in our analysis herein. To the extent the parties disagree regarding the application of a POSA's general knowledge of nucleic acids and related prior art techniques to the Petition's grounds, such disputes are addressed in our analysis below.

C. Claim Construction

The parties assert that all of the claim terms have their plain and ordinary meaning and that no formal claim construction is necessary. *See* Pet. 17; Resp. 24. Nevertheless, both sides accuse the other of misconstruing the "gRNA functionality" recited in the challenged claims. *See* Resp. 24–25;

Reply 16; Sur-reply 2–4. Accordingly, we begin by briefly clarifying the gRNA functionality required by the claims.

Independent claims 1 and 19 recite a guide RNA or crRNA molecule having "gRNA functionality comprising associating with a Cas protein and targeting the gRNA:Cas protein complex to the target sequence." Ex. 1001, 257:42–44, 259:5–7. This claim language makes clear that the recited "gRNA functionality" requires a molecule that can: (1) associate with a Cas protein, and (2) target that complex to a target sequence in a polynucleotide. To the extent Petitioner suggests that a gRNA "need *only* bind to the Cas protein" to have "gRNA functionality," we disagree because the claims additionally recite the above-quoted targeting functionality. *See* Reply 16 (emphasis added).

At the same time, we agree with Petitioner that "a gRNA need not cleave the target DNA" to have the claimed "gRNA functionality." Reply 16. The Specification defines "gRNA functionality" as "one or more functions of naturally occurring guide RNA, such as associating with a Cas protein, or a function performed by the guide RNA in association with a Cas protein" and lists various gRNA functions, e.g., "associating," "targeting," "binding," "nicking," and "cleaving," that may be present "in certain embodiments" of the invention. Ex. 1001, 6:56–7:1. However, as noted above, the claims recite "gRNA functionality comprising" only the associating and targeting functions. Thus, while a molecule may have additional functions such as cleaving a target polynucleotide, it need only exhibit the recited "associating" and "targeting" functions to have the "gRNA functionality" in claims 1 and 19.

Our determination that "gRNA functionality" requires the recited "associating" and "targeting" functions, but not additional, unrecited functions such as cleaving, is also consistent with the parties' position that the claim language has its plain and ordinary meaning. *See* Pet. 17; Resp. 24. The parties do not dispute the construction of any other terms, nor do we discern that any further claim construction is necessary to resolve the issues in this proceeding. *See Nidec Motor Corp. v. Zhongshan Broad OceanMotor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (explaining that it is only necessary to "construe terms 'that are in controversy, and only to the extent necessary to resolve the controversy'" (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))).

D. Overview of the Cited References

i. Pioneer Hi-Bred

Pioneer Hi-Bred is a publication of a PCT application filed August 20, 2014. Ex. 1006, code (22). Patent Owner does not dispute Petitioner's assertion that Pioneer Hi-Bred is prior art to the challenged claims. Pet. 14.

Pioneer Hi-Bred describes "methods and compositions employ[ing] a guide polynucleotide/Cas endonuclease system to provide an effective system for modifying or altering target sites within the genome of a cell or organism." Ex. 1006, Abstr. Pioneer Hi-Bred defines the term "guide polynucleotide" as "a polynucleotide sequence that can form a complex with a Cas endonuclease and enables the Cas endonuclease to recognize and optionally cleave a DNA target site." *Id.* at 24:6–8.[8] Pioneer Hi-Bred teaches that the polynucleotide "can be a single molecule or a double molecule" and that "[a] guide polynucleotide that solely comprises ribonucleic acids is also referred to as a 'guide RNA.'" *Id.* at 24:9–20.

Pioneer Hi-Bred discloses a guide RNA with a variable targeting domain (VT domain) having a 3'-end "that is complementary to a nucleotide sequence in a target DNA" and a Cas endonuclease recognition domain (CER domain) having a 5'-end "that interacts with a Cas endonuclease." Ex. 1006, 24:21–25: 28, Fig. 1A–1B (depicting single and duplex guide polynucleotides). Pioneer Hi-Bred explains that "[t]he VT domain is responsible for interacting with the DNA target site through direct nucleotide-nucleotide base pairings while the CER domain is required for proper Cas endonuclease recognition (Figure 3A and Figure 3B)." *Id.* at 105:5–8. Pioneer Hi-Bred teaches that these domains in the guide polynucleotide "function to link DNA target site recognition with Cas endonuclease target site cleavage." *Id.* at 105:9–11; *see also id.* at Fig. 3A-3B (depicting complexes formed between a single and duplex guide RNA and a Cas9 endonuclease).

Pioneer Hi-Bred also discloses that the guide polynucleotide may contain "synthetic, non-natural, or altered nucleotide bases" as well as other modifications such as "a fluorescent label." Ex. 1006, 27:3–19, 61:19–20. In Example 4, Pioneer Hi-Bred describes "modifying the nucleotide base, phosphodiester bond linkage or molecular topography of the guiding nucleic

[8] Unless otherwise indicated, the pinpoint cites in this decision refer to the page number in the original document as opposed to the number in the exhibit label.

acid component(s) of the guide polynucleotide/Cas endonuclease system." *Id.* at 104:15–105:2. Table 7 of Example 4 provides "[e]xamples of nuclease resistant nucleotide and phosphodiester bond modifications," including "2'-O-Methyl RNA Bases" and "Phosphorothioate bond[s]," that may be introduced in order "to reduce unwanted degradation" of the guide polynucleotide. *Id*. at 106:13–107:5. Pioneer Hi-Bred discloses that

> [m]odifications may be introduced at the 5' and 3' ends of any one of the nucleic acid residues comprising the VT or CER domains to inhibit exonuclease cleavage activity, can be introduced in the middle of the nucleic acid sequence comprising the VT or CER domains to slow endonuclease cleavage activity or can be introduced throughout the nucleic acid sequences comprising the VT or CER domains to provide protection from both exo- and endo-nucleases.

Id. at 106:19–25. According to Pioneer Hi-Bred, these modified guide polynucleotides may be used "in any organism subject to genome modification with the guide polynucleotide/Cas endonuclease system." *Id.* at 108:3–5.

In Example 5 of Pioneer Hi-Bred, "some the nucleotide base and phosphodiester bond modifications described in Example 4 are introduced into the VT domain and/or CER domain of a crNucleotide." Ex. 1006, 108:16–18. Table 8 of Example 5, reproduced in part below, describes crRNA sequences with modifications, including modifications "near ends" or "at ends" of the VT and CER domains (i.e., sequences 64–69).

Table 8. crRNA and crDNA nucleotide base and phosphodiester linkage modifications.

Nucleic Acid Type	Modification	crRNA or crDNA Sequence and Corresponding Modification[1]	
		VT Domain	CER Domain
crRNA	None	GCGUACGCGUACGUGUG (SEQ ID NO: 62)	GUUUUAGAGCUAUGCUGUUUUG (SEQ ID NO: 63)
crRNA	Phosphorothioate bonds near ends	G*C*G*UACGCGUACGUGUG (SEQ ID NO: 64)	GUUUUAGAGCUAUGCUGUU*U*U*G (SEQ ID NO: 65)
crRNA	2'-O-Methyl RNA nucleotides at ends	mGmCmGUACGCGUACGUGUG (SEQ ID NO: 66)	GUUUUAGAGCUAUGCUGUUmUmUmG (SEQ ID NO: 67)
crRNA	2'-O-Methyl RNA nucleotides for each nucleotide	mGmCmGmUmAmCmGmCmGmUmAmCmGmUmGmUmG (SEQ ID NO: 68)	mGmUmUmUmUmAmGmAmGmCmUmAmUmGmCmUmGmUmUmUmUmG (SEQ ID NO: 69)

Id. at 109. The excerpt from Table 8 above shows modifications comprising phosphorothioate bonds (denoted with a "*") and 2'-O-methyl RNA nucleotides (denoted with a "m") to particular nucleotides of a crRNA sequence. *See id.* at 109–110, n.1. The first nucleotide in the sequences in the VT Domain column is at the 5' end of the crRNA and the last nucleotide in the sequences in the CER Domain column is at the 3' end. *See* Ex. 1006, Fig. 1A; Ex. 1002 ¶¶ 126–29, 131–34. Thus, for example, sequence 66 discloses a crRNA with 2'-O-methyl modifications to the sugars of the three nucleotides at the 5' end and sequence 67 discloses a crRNA with 2'-O-methyl modifications to the sugars of the three nucleotides at the 3' end.

Pioneer Hi-Bred discloses that modifications "similar to those illustrated in Example 5 Table 8 can be introduced individually or in combination into the crRNA, crDNA, tracrRNA, tracrDNA, long guide RNA or long guide DNA nucleic acid components of the guide polynuclease system and synthesized per standard techniques." Ex. 1006, 113:25–29.

ii. Krutzfeldt

Krutzfeldt was published in 2007. Ex. 1009. Patent Owner does not dispute that Krutzfedlt is prior art to the challenged claims.

According to Krutzfeldt, "MicroRNAs (miRNAs) are an abundant class of 20-23-nt long regulators of gene expression." Ex. 1009, Abstr. Krutzfeldt describes analogs of these miRNAs referred to as "antagomirs." *Id.* at 2885. These antagomirs "differ from normal RNA by complete 2'-*O*-methylation of sugar, phosphorothioate backbone and a cholesterol-moiety at 3'-end." *Id.* at 2885. Krutzfeldt discloses the combination of 2'-O-methyl and phosphorothioate modifications on nucleotides at the 5' and 3' ends of its antagomirs. *Id.* at 2886 (Table 1 disclosing antagomir sequences with a lower case letters indicating a 2'-O-methyl modification and a superscript indicating a phosphorothioate linkage). Krutzfeldt teaches these modifications "protect against different RNAse activities" that would otherwise degrade the RNA strand. *See id.* at 2889 (teaching "phosphorothioate modification to protect against exonucleases" and 2'-O-methyl sugar modification "to protect against endonuclease activity").

iii. Deleavey

Deleavey is a review article published in 2012. Ex. 1007. Patent Owner does not dispute that Deleavey is prior art to the challenged claims.

Deleavey teaches that oligonucleotides (ONs) such as small interfering (siRNAs) and microRNA-targeting ONs (anti-miRNAs) "and their chemically modified mimics, are now routinely used in the laboratory" and "under active investigation in the clinic." Ex. 1007, 937. Deleavey teaches that an array of ON chemical modifications have been developed to overcome the "therapeutically limiting features" of RNAs. *Id.*

In particular, Deleavey explains that RNAs "are rapidly degraded in cells . . . leading to shortened duration of activity and systemic delivery challenges." Ex. 1007, 941. Deleavey teaches that chemical modifications to both the internucleotide linkage, e.g., phosphorothioate and phosphonoacetate (PACE), and sugar, e.g., 2'-O-methyl nucleosides, had been shown to improve stability in various RNA applications. *See id.* at 942–47.

iv. Soutschek

Soutschek was published in 2004. Ex. 1012, 173. Patent Owner does not dispute that Soutschek is prior art to the challenged claims.

Soutschek describes chemically modified short interfering RNAs (siRNAs). Ex. 1012, 173. Soutschek teaches that "[c]hemically stabilized siRNAs with partial phosphorothioate backbone and 2'-O-methyl sugar modifications on the sense and antisense strands showed significantly enhanced resistance towards degradation by exo- and endonucleases in serum and tissue homogenates." *Id.*; *see also id.* at 177 (listing sequences of chemically-modified siRNAs with phosphorothioate and 2'-O-methyl sugar modifications on certain nucleotides).

v. Yoo

Yoo was published in 2004. Ex. 1011, 2008. Patent Owner does not dispute that Yoo is prior art to the challenged claims.

Yoo describes antisense oligodeoxynucleotides (ODNs) having both phosphorothioate and 2'-O-methyl sugar modifications. Ex. 1011, 2008.

Yoo teaches that "the addition of 2'-O-methyl groups to a phosphorothioate-modified ODN is advantageous because of increased stability of binding and reduced non-specific effects." *Id.*

vi. Threlfall

Threlfall was published in 2012. Ex. 1010, 746. Patent Owner does not dispute that Threlfall is prior art to the challenged claims.

Threlfall describes "[c]himeric 2'-O-methyl oligoribonucleotides (2'-OMe ORNs) containing internucleotide linkages which were modified with phosphonoacetate (PACE) or thiophosphonoacetate (thioPACE)" at their ends. Ex. 1010, 746; *see also id.* at 747 (Table 1 showing chemically modified sequences). Threlfall explains that "[o]ligoribonucleotides with a 2'-O-methyl modification . . . are known to be nuclease resistant and increase the stability of a duplex which is formed with complementary RNA." *Id.* at 746. Moreover, Threlfall teaches that "ODNs modified with PACE or thioPACE [had been] shown to be nuclease resistant" in a prior study. *Id.* at 747. Threlfall reports results from tests on ORNs combining these chemical modifications "into chimeric 2'-OMe ORNs as PACE or thioPACE modifications." *Id.* at 752. According to Threlfall, "the chimeric ORNs formed stable duplexes with complementary RNA, and the majority of these duplexes had higher thermal melting temperatures than an unmodified RNA:RNA control duplex." *Id.*

E. Ground 1: Anticipation by Pioneer Hi-Bred

Petitioner contends that claims 1–5, 8–21, and 21–33 are anticipated by Pioneer Hi-Bred. *See* Pet. 17–47. As explained below, Petitioner has shown by a preponderance of the evidence that these claims are anticipated by Pioneer Hi-Bred.

i. Claims 1 and 19

Petitioner has shown that Pioneer Hi-Bred discloses guide RNA and crRNA molecules having 2'-O-methyl modifications as recited in claims 1 and 19. *See, e.g.*, Pet. 17–30, 44 (showing for claims 1 and 19). In particular, Table 8 in Pioneer Hi-Bred discloses exemplary crRNA molecules comprising 2'-O-methyl modifications in the VT domain (e.g., sequences 66 and 68). Ex. 1006, 109. The VT domain corresponds to the "guide sequence" in claims 1 and 19 because it is "complementary to a nucleotide sequence in a target DNA" meaning that it hybridizes to a target sequence in that polynucleotide. *Id.* at 24:23–28, 97:17–18, Fig. 3A–B; Ex. 1003 ¶¶ 130–33. Pioneer Hi-Bred teaches that crRNAs, including those in Table 8, also have a CER domain (i.e., a stem sequence) that is complementary to and paired with a tracrRNA to form a duplex guide RNA. *Id.* at 8:22–31, 24:32–25:10, 109:4–9, Fig. 1A, 3A; Ex. 1003 ¶¶ 129–33, 138. Pioneer Hi-Bred further discloses embodiments in which modifications similar to those exemplified in Table 8 are introduced into a long guide RNA (i.e., a single guide RNA) in which the crRNA and tracrRNA are fused together. *Id.* at 113:25–29; *see also id.* at 10:8–13, Fig. 1B, 3B (depicting a "Long guide RNA" as a "fused crRNA and tracrRNA")); Ex. 1003 ¶¶ 125–29, 135–37.

Petitioner has also shown that Pioneer Hi-Bred discloses the recited "gRNA functionality." *See* Pet. 25–26 (Ex. 1003 ¶¶ 128–29, 142–46). Pioneer Hi-Bred defines the term "guide polynucleotide" as "a polynucleotide sequence that can form a complex with a Cas endonuclease and enables the Cas endonuclease to recognize and optionally cleave a DNA target site." Ex. 1006, 24:6–8. Thus, Pioneer Hi-Bred discloses that the guide polynucleotides described therein can: (1) form a complex with a Cas endonuclease; and (2) enable the endonuclease to recognize a DNA target site. That disclosure reads on both the associating and targeting aspects of the "gRNA functionality" recited in claims 1 and 19.

Pioneer Hi-Bred refers to the modified sequences in Examples 4 and 5 as "modified guide polynucleotides," indicating that those sequences have this functionality. Ex. 1006, 107:14–18:11; *see also id.* at 109:4–6 (referring to the complex formed by the "modified crRNA or crDNA components described in Table 8" as a "modified guide polynucleotide/Cas endonuclease complex"). Other statements in these examples confirm this understanding. *See id.* at 107:14–24 (explaining that modified guide polynucleotides may be delivered with the other components of the "guide polynucleotide/Cas endonuclease system" to "form a functional complex capable of binding and/or cleaving a chromosomal DNA target site"); 107:24–108:2 ("Modified guide polynucleotides described above may also be delivered simultaneously in multiplex to target multiple chromosomal DNA sequences for cleavage or nicking.").[9]

[9] While the claimed "gRNA functionality" does not require cleavage, the fact that cleavage occurs at a target site indicates that a gRNA is capable of associating with a Cas endonuclease and targeting it to a particular site. *See* Ex. 1059 ¶ 31 (explaining that "Agilent's cleavage experiments [in the Specification of the '034 patent] that show cleavage activity for certain gRNAs establish that 'gRNA functionality' is present for gRNAs," but the absence of cleavage activity "does not mean that it does not have 'gRNA functionality' because such a gRNA may nonetheless form a complex with Cas without effecting target cleavage").

Accordingly, we agree with Petitioner that Pioneer Hi-Bred discloses synthetic guide RNA and crRNA molecules as recited in claims 1 and 19. In its Response, Patent Owner contends that Pioneer Hi-Bred does not disclose the recited "gRNA functionality" and is non-enabling. *See* Resp. 29–44. As explained below, both of these arguments are unavailing.

1. Whether Pioneer Hi-Bred discloses a functional gRNA

Patent Owner argues that "Pioneer Hi-Bred does not disclose ***a single functional modified gRNA***." Resp. 30. According to Patent Owner, Pioneer Hi-Bred tested only a single modified guide polynucleotide made of DNA, which subsequent testing revealed was not successful. *See id.* at 30–34. Patent Owner argues that Examples 4 and 5 "do not reveal any actual testing" and are "simply an invitation to experiment . . . that precludes a finding of anticipation because functionality is not disclosed." *Id.* at 34–36.

Patent Owner argues that Petitioner did not assert "anticipation by inherency" for the gRNA functionality limitation. Resp. 36. According to Patent Owner, this is because "some of the sequences identified in Table 8 exhibited no functionality, including some 2'-O-Methyl modifications, so it cannot be assumed that this combination will work." *Id.* Patent Owner points to results in the Specification of the challenged patent showing that the use of chemically modified gRNAs with 26 and 37 consecutive 2'-O-methyl-modified nucleotides at the 5' end did not result in cleavage activity. *Id.* at 37 (referring to Table 4 results for entry nos. 151 and 152 in Ex. 1001). According to Patent Owner, these results show "the proposed modifications in sequences 68 and 69 [in Table 8] together are non-functional." *Id.*

Patent Owner further contends that "a POSA would doubt" that sequences 64–69[10] in Table 8 of Pioneer Hi-Bred "would function at all because each design is both truncated and is also modified." Sur-reply 8 (citing Ex. 2025 ¶¶ 201–03, 221, 242–46); Ex. 1026, Ex. 1033, Ex. 2044, Ex. 2045). Patent Owner's expert, Dr. Marshall testifies that "a putative guide sequence is 20 nt in length," whereas the VT domain sequences in Pioneer Hi-Bred's sequences 64, 66, and 68 are "each only 17 nt long." Ex. 2025 ¶ 201. According to Patent Owner and Dr. Marshall, this means "the nucleotides being edited in Pioneer Hi-Bred are 4, 5, and 6 because 1, 2, and 3 have already been truncated from the end. But Nishimasu found that positions 4, 5, and 6 were crucial positions in the context of gRNA associating and targeting." Resp. 38–39; *see also* Ex. 2025 ¶¶ 202–03.

Petitioner replies, urging that testing data is not required for a prior art reference to be anticipatory. *See* Reply 2–5 (citing cases). Petitioner argues that Pioneer Hi-Bred provides a targeted disclosure that identifies only five types of modifications for decreasing unwanted nuclease degradation and "explains precisely why those modifications should be used." *Id.* at 13–14 (citing Ex. 1006, 107 (Table 7)). Moreover, Petitioner urges that Pioneer Hi-Bred "presents exemplary embodiments of gRNAs [in sequences 64–69 of Table 8] containing those modifications and the precise positions where those modifications ought to be placed, including the types and locations of

[10] Sequences 64 and 65 disclose phosphorothioate modifications. While not recited in claims 1 and 19, such modifications are recited in some of the dependent claims. *See* Ex. 1001, 257:62–64, 258:41–57, 259:15–17, 259:24–260:10 (claims 5, 9–13, 21, and 24–28). We address them here for completeness.

modifications that anticipate" the challenged claims. *Id.* at 14–15 (citing Ex. 1006, 109).

Regarding Patent Owner's argument that the cleavage data in the '034 patent Specification shows that Pioneer Hi-Bred's sequences 68 and 69 lack functionality, Petitioner points out that the "gRNA functionality" in claims 1 and 19 does not require cleavage. Reply 16–17. Thus, Petitioner urges there is not "a shred of evidence that the sequences 68 and 69 in Pioneer Hi-Bred do not show the type of 'gRNA functionality'" in claims 1 and 19. *Id.* at 17 (citing Ex. 1060, 129:17–23; Ex. 1061, 105:15–25, 106:16–24, 107:14–25, 108:13–109:11).

Petitioner replies to Patent Owner's truncation argument and the related testimony of Dr. Marshall with evidence from Dr. Furneaux. *See* Ex. 1059 ¶¶ 73–75. According to Dr. Furneaux, the record does not support Dr. Marshall's assumption that the sequences in Table 8 are truncated because Pioneer Hi-Bred teaches that the VT domain varies in length and is "designed based on the target sequence," which for the crRNAs in Table 8 is the 17 nucleotides long. *Id.* ¶ 74. Dr. Furneaux further testifies that the references Dr. Marshall cites as showing that truncated gRNAs do not work actually "support the opposite conclusion, that gRNAs with target sequences of 17 nucleotides are functional." *Id.* ¶ 75.

We find Petitioner's arguments and evidence persuasive. It is well established that "anticipation does not require actual performance of suggestions in a disclosure." *Novo Nordisk Pharms., Inc. v. Bio-Tech. Gen. Corp.*, 424 F.3d 1347, 1355 (Fed. Cir. 2005); *see also In re Donohue*, 766 F.2d 531, 533 (Fed. Cir. 1985). Moreover, the Petition need not assert that "gRNA functionality" is inherently disclosed because Petitioner has shown that Pioneer Hi-Bred's expressly discloses this limitation. Pet. 25–26 (citing evidence). There is no additional requirement that this express disclosure be backed by test data in order for the reference to be anticipatory. *See, e.g.*, *Novo*, 424 F.3d at 1355; *Rasmusson v. SmithKline Beecham Corp.*, 413 F.3d 1318, 1326 (Fed. Cir. 2005) ("[P]roof of efficacy is not required in order for a reference to be enabled for purposes of anticipation.").

Patent Owner suggests we should ignore what it calls a "bald assertion" of gRNA functionality in Pioneer Hi-Bred because either that assertion is incorrect or a POSA would doubt that it was true. *See* Resp. 2. We disagree. To begin with, Patent Owner's reliance on the purported failure of Pioneer Hi-Bred's DNA-based embodiments is unavailing. Pioneer Hi-Bred discloses both DNA and RNA-based embodiments. The Petition is premised on the latter. Even if we accept Patent Owner's argument that the DNA-based examples lack gRNA functionality, that fact does not suggest that a POSA would doubt that the RNA-based embodiments, e.g., crRNAs comprising sequences 64–69 in Table 8, lack such functionality.

Patent Owner's argument that a POSA would doubt whether Pioneer Hi-Bred's RNA-based embodiments have gRNA functionality because the VT domain in sequences 64, 66, and 68 is truncated is also unpersuasive. First, Pioneer Hi-Bred does not state that these sequences are truncated, nor does the record support Patent Owner's argument that these sequences are necessarily truncated because they are 17 nt, as opposed to 20 nt, long. To the contrary, Pioneer Hi-Bred teaches that the VT domain may vary in length from 12 to 30 nt. Ex. 1006, 3:19–20, 98:19–23. In addition, Pioneer Hi-Bred explains that the sequences in Table 8 are designed to target the "LIGCas-3" site in Maize, which is 17 nt long. *Id.* at 109:4–9; *see also id.* at

99:15–18 (Table 1 identifying 17 nt target site sequence for LIGCas-3). Thus, the record supports, and we credit, Dr. Furneaux's testimony that the VT domain for these sequences is 17 nt because it is designed to target a 17 nt sequence—not because it has been truncated.[11] Ex. 1059 ¶ 74.

Second, even if those sequences were truncated, the record does not support Patent Owner's argument that a POSA would doubt that they have gRNA functionality. Patent Owner's argument is premised on Dr. Marshall's testimony that these sequences "are truncated to a point that would render them nonfunctional." Ex. 2025 ¶ 200. But the references he cites do not support that position. *See* Ex. 2025 ¶ 202 (citing Ex. 1033 ("Fu"); Ex. 2044 ("Cencic"); and Ex. 2045 ("Mali")). To the contrary, Fu evidences that "truncated gRNAs, with shorter regions of target complementarity <20 nucleotides in length" are functional and that truncation to 17 nt is beneficial. Ex. 1033, Abstr.; *see also id.* at 283 ("[W]e found that [truncated] gRNAs with 17 or 18 nucleotides of complementarity generally function efficiently at the intended target site and have improved specificities."). Cencic and Mali likewise do not suggest that a 17 nt VT domain would lack gRNA functionality. *See* Ex. 2025 ¶ 202 (citing Cencic as showing "truncated gRNA tests showed that a guide sequence of only 16 nucleotides" abolished cleavage activity); Ex. 2045, Supp. Fig. 10 ("1-3 bp 5' [gRNA]

[11] Patent Owner asserts that Dr. Furneaux "adopted the definition of 'truncated gRNAs' as those with 'regions of target complementarity <20 nucleotides in length" in paragraph 79 of his opening declaration. Sur-reply 8 n. 1. We disagree. The testimony in paragraph 79 refers to the particular results in Fu. *See* Ex. 1003 ¶ 78 (citing Ex. 1033). We see nothing there that suggests Dr. Furneaux agrees with Patent Owner's position that any gRNA with a VT region of less than 20 nt is truncated.

truncations are indeed well tolerated, but larger deletions lead to loss of activity”). Accordingly, we credit Dr. Furneaux’s testimony (*see, e.g.*, Ex. 1003 ¶ 78, Ex. 1059 ¶¶ 73–75) over the competing testimony offered by Dr. Marshall on these points.

At oral argument, Patent Owner took its argument a step further asserting not only that a POSA would doubt their functionality, but also that that the record shows that Pioneer Hi-Bred sequences 64–67, in fact, lack gRNA functionality because they combine truncation with chemical modifications to the 4, 5, and 6 nucleotides of the untruncated gRNA. *See* Tr. 41:3–48:16 (urging that “Nishimasu” (Ex. 1026) shows that the nucleotides at this position are critical to gRNA functionality). This new argument is unavailing.[12] First, as explained above, the record does not support Patent Owner’s argument that the VT domain in these sequences is truncated. Second, there is test data in the record demonstrating gRNA functionality for both truncated gRNAs and gRNAs with modifications to the 4, 5, and 6 nucleotides. *See, e.g.*, Ex. 1033, 283; Ex. 2045, Supp. Fig. 10;

[12] In its papers, Patent Owner asserts that a POSA would “doubt” the gRNA functionality of sequences 64–69 for this reason. Sur-reply 8; *see also* Resp. 39 (arguing that the “combination of truncation and modifications would have been unpredictable” and thus gRNA functionality was not “inherent”); Ex. 2025 ¶ 203 (“[A] POSA would seriously question whether the guide sequence designs of Table 8 would be functional”). But at the hearing, Patent Owner stated it was asserting “both” that a POSA would doubt their functionality and that these sequences do not, in fact, exhibit gRNA functionality and that the record demonstrates such. Tr. 41:3–16. This latter point is a new argument that was not clearly presented in the Response and has therefore been waived. Even if it had been timely presented, we find that the record does not support Patent Owner’s position that Pioneer Hi-Bred’s sequences 64–69 are non-functional.

Ex. 1001, Tables 3 and 4 (Patent Owner's Specification showing cleavage activity for truncated gRNAs with a 17 nt target sequence and gRNAs with chemical modifications at the 4, 5, and 6 nucleotides). While not prior art, this data does tend to rebut Patent Owner's argument that sequences 64–67 lack gRNA functionality because it shows that both truncation to 17 nt and modifications to the 4, 5, and 6 nucleotides can be individually tolerated without losing cleavage activity.

We also disagree with Patent Owner's argument that the cleavage data in its Specification shows that a crRNA corresponding to Pioneer Hi-Bred's sequences 68 and 69 would lack "gRNA functionality." Resp. 36–37. The data Patent Owner points to shows only that cleavage did not occur. Cleavage, however, is not required for the "gRNA functionality" recited in claims 1 and 12. Patent Owner's expert, Dr. Sampson admitted on cross-examination that just because a gRNA in Table 4 lacks cleavage activity does not demonstrate that it also lacks the ability to bind a Cas protein and target that complex to target polynucleotide:

> Q. [T]he handful of guide RNAs that you identified in Table 4 of the ['034] patent as not exhibiting cleavage, those may nonetheless bind with a Cas protein and form a complex with the target DNA; correct?
> A. Yeah. I can't, you known, answer that definitively.
> Q. All we know is that it might or might not happen; right? We don't know one way or the other?
> A. There is no objective evidence that would be able to indicate that.
>
> ***
>
> Q. I just want to be clear, the handful of guide RNAs that you point out as being nonfunctional, you don't actually know

> whether they actually have gRNA functionality as claimed in the claims; right?
> A. What I'd say is that I can't – there's no objective evidence that allows you for a determination of whether the guides bind to Cas9 or direct the programmed Cas9 to the target site.
>
> ***
>
> Q. You can't say one way or another whether the guide RNAs that you say are not functional actually lack that gRNA functionality as claimed in the claims; right?
> A. I can state that I have no evidence to differentiate between whether they would bind to the Cas or direct the Cas to a target sequence.

Ex. 1060, 106:16–109:11 (objections omitted); *see also* Ex. 1059 ¶ 31 (similar testimony from Dr. Furneaux). Thus, the data in Table 4 of the Specification showing a lack of cleavage activity does not demonstrate that the corresponding gRNA lacks the claimed "gRNA functionality."

Moreover, Patent Owner's argument only applies to a crRNA that combines sequences 68 and 69 "together" resulting in a crRNA with 39 consecutive 2'-O-methyl modifications. Resp. 37. However, Pioneer Hi-Bred discloses that these modifications may be introduced "individually or in combination." Ex. 1006, 113:25–29. If introduced individually, a crRNA having the modifications in sequence 68 would have 17 2'-O-methyl modifications and a crRNA having the modifications in sequence 69 would have 22 such modifications. This is closer to the number of modifications in sequences that the data in the Specification shows do have cleavage activity (e.g., Table 4 Entry #s 146–150 having 20 consecutive 2'-O-methyl modifications) than it is to the sequences Patent Owner identifies as lacking cleavage activity. *See* Resp. 37 (citing Ex. 1001, 119 (Table 4 Entry #s 151 and 152 having 26 and 37 consecutive sugar modifications). Therefore, the data in the Specification does not show that a crRNA comprising sequence 68 or 69 would lack cleavage activity, much less the broader "gRNA functionality" recited in the challenged claims.

For these reasons, we determine that Pioneer Hi-Bred discloses functional, chemically-modified gRNAs as recited in claims 1 and 19.

2. Whether Pioneer Hi-Bred is enabled

Patent Owner contends that Pioneer Hi-Bred is not enabled. Resp. 39–44. More specifically, Patent Owner asserts that "[a] POSA encountering Pioneer Hi-Bred could not make the claimed inventions of the '034 [patent] without undue experimentation" because it "discloses a laundry list of chemical modifications, that can be made alone or in combination, and applied literally anywhere in the disclosed guides." *Id.* at 43. According to Patent Owner, the "art was new, complicated and unpredictable," "not 'mechanistically analogous' to anything that came before it," and "a POSA would have been circumspect about making the claimed modifications without testing because what was known about the nature of the interactions between the Cas Protein and the gRNA suggested the gRNA would be very sensitive to modifications, especially in the guide portion." *Id*. at 44 (no citations provided). Thus, argues Patent Owner, "the *Wands* factors lean[] heavily in favor of a finding that Pioneer Hi-Bred would not enable one to make the claims of the '034 Patent without undue experimentation." *Id.* (italics added).

Petitioner contends that the anticipating disclosures in Pioneer Hi-Bred are enabled. Reply 5–18. In particular, Petitioner urges that Pioneer Hi-Bred discloses only five types of chemical modifications for decreasing unwanted nuclease degradation in Table 7 and provides exemplary embodiments of gRNAs, including those having "the types and locations of modifications that anticipate the '034 Patent claims," in Table 8. *Id.* at 13–15. Petitioner points to testimony from Patent Owner's declarants showing that the techniques for making such gRNAs were known in the art and a POSA could use "commercially available instruments that could churn out multiple [chemically-modified] gRNAs in a single day." *Id.* at 8–9.

Petitioner also cites evidence that the chemical modifications disclosed in Pioneer Hi-Bred had been used for decades prior to the filing of the '034 patent "to stabilize RNA against nucleases" and that "CRISPR gRNA stabilization presented a new iteration of an old problem with a tried and true solution." *Id.* at 9–12. According to Petitioner, "the testing data in the patent drives home just how predictable this field was" because only 7 of the roughly 250 gRNAs Patent Owner tested lacked cleavage functionality. *Id.* at 12–13, 16 (referring to the inventors "roughly 97% success rate in predicting which modified gRNAs would work" based on prior art teachings).

We again find Petitioner's evidence and argument persuasive. For a prior art reference to be enabling, "the reference need only enable a single embodiment of the claim." *In re Morsa*, 803 F.3d 1374, 1377 (Fed. Cir. 2015). "In other words, a prior art reference need not enable its full disclosure; it only needs to enable the portions of its disclosure alleged to anticipate the claimed invention." *In re Antor Media Corp.*, 689 F.3d 1282, 1290 (Fed. Cir. 2012). Here, Petitioner asserts that the RNA-based embodiments disclosed in Examples 4 and 5 of Pioneer Hi-Bred are anticipatory. Those disclosures are presumed enabling and Patent Owner has not shown otherwise.

Indeed, the record demonstrates that a POSA, as of December 2014, could practice these disclosures without undue experimentation.[13] As explained above, Table 7 of Pioneer Hi-Bred teaches modifications, including 2'-O-methyl modifications to the sugar and phosphorothioate bond modifications, that can be used to decrease unwanted nuclease degradation of a gRNA in a guide polynucleotide/Cas endonuclease system. And Table 8 discloses exemplary sequences of such chemically-modified crRNAs that read on claims 1 and 12. Pioneer Hi-Bred teaches that gRNAs having the modifications in Table 8 and similar modifications can be "synthesized per standard techniques." Ex. 1006, 113:25–29.

While Pioneer Hi-Bred does not further describe these standard techniques, the record demonstrates such techniques were known in the art and a POSA would have been able to use them to make the gRNAs disclosed in Pioneer Hi-Bred without undue experimentation. Both sides' experts testify that techniques such as "click chemistry" and "TC chemistry" were known in the prior art for synthesizing long oligonucleotides. Ex. 1059 ¶¶ 33–38; *see* Ex. 1061, 188:9–190:17). Moreover, the Specification of the '034 patent cites references describing these known techniques to explain

[13] The question of undue experimentation involves consideration of certain factors identified in *In re Wands*, 858 F.2d 731, 737 (Fed. Cir. 1988). These factors include: the quantity of experimentation, the amount of direction or guidance present, the presence or absence of working examples, the state of the prior art, the relative skill of those in the art, and the predictability or unpredictability of the art. *Id.* To the extent they are applicable here, we have considered these and the other *Wands* factors in our analysis.

how its chemically-modified gRNAs may be synthesized. Ex. 1001, 48:4–33; *see also In re Morsa*, 803 F.3d at 1378 (explaining that statements in the specification evidencing the knowledge of a POSA may be relied upon to show that a prior art reference is enabled). The record further evidences that the inventors were able to employ such techniques to make an individual gRNA in about a day's time and that commercially-available "synthesizers" were available, allowing dozens of different gRNAs to be synthesized at the same time. Ex. 1069 ¶ 41; *see* Ex. 1060, 87:6–18, 88:3–16 (Dr. Sampson testifying that synthesizers capable of making as many as 48 gRNAs at a time were commercially-available), 191:16–192:14 (testifying that multiple guides can be made in a day).

Patent Owner argues that its inventors were uniquely-skilled in the synthesis of long RNAs and thus "it would have been extremely challenging for a POSA to chemically synthesize the claimed chemically-modified gRNA." *See* Resp. 11–13. This argument is unavailing for several reasons. First, the Specification of the '034 patent does not disclose any new techniques for synthesizing chemically-modified gRNAs, but instead refers to the use of click chemistry and TC chemistry techniques already taught in other references. Ex. 1001, 48:4–33. Second, while Dr. Sampson points to challenges he and the other inventors allegedly had to overcome to make these molecules (*see* Ex. 2029 ¶¶ 12–15), none of those challenges are mentioned in the Specification of the '034 patent. *See In re Epstein*, 32 F.3d 1559, 1568 (Fed. Cir. 1994) (holding "the Board's observation that appellant did not provide the type of detail in his specification that he now argues is necessary in prior art references supports the Board's finding that one skilled in the art would have known how to implement the features of the references”). Moreover, when cross-examined about these purported challenges, Dr. Sampson was either unable to remember how the inventors addressed them or he testified they were overcome by increasing the reaction time, which was “[j]ust kind of a pretty standard thing that you do.” Ex.1061, 115:3–118:19. Third, in order to anticipate, Pioneer Hi-Bred “need only enable a single embodiment of the claim.” *In re Morsa*, 803 F.3d at 1377. Petitioner’s arguments that its team of inventors spent over a year to synthesize “hundreds” of gRNAs does not show that the experimentation required to make any one of the anticipating gRNAs disclosed in Pioneer Hi-Bred, e.g., the exemplary crRNAs in Table 8, would be undue. For these reasons, we credit Dr. Furneaux’s testimony that undue experimentation would not be required to make the anticipating, chemically-modified gRNAs taught in Pioneer Hi-Bred (*see* Ex. 1059 ¶¶ 32–47) over the competing testimony of Patent Owner’s declarants.

So too, the fact that Pioneer Hi-Bred “contains no data regarding any testing of the sequences in Table 8” does not demonstrate that the crRNAs disclosed there are not enabled. *See* Resp. 34–36. “It is not . . . necessary that an invention disclosed in a publication shall have actually been made in order to satisfy the enablement requirement.” *In re Donohue*, 766 F.2d 531, 533 (Fed.Cir.1985). Thus, while it appears that Examples 4 and 5 in Pioneer Hi-Bred are prophetic, as opposed to working, examples, that fact alone does not undermine the presumption that Pioneer Hi-Bred is enabled. *See Antor Media,* 689 F.3d at 1289–90 (“[T]he mere use of forward-looking language (such as terms like ‘should’) does not show one way or another whether a person of ordinary skill in the art would have to engage in undue experimentation to perform the claimed invention”).

To the extent Patent Owner contends that the nascent state of the art demonstrates that undue experimentation would be required, we disagree. *See* Resp. 44. It is undisputed that the use of gRNA in a CRISPR/Cas system was a relatively new discovery first published in mid-2012. *See* Pet. 2; Resp. 52–53. That said, the record demonstrates that by December 2014 substantial research into such systems had been published and would have been known to a POSA Ex. 1003 ¶ 78; Ex. 1059 ¶¶ 20–23, 65–72 (citing references). A POSA would also know that gRNA was subject to degradation, which could limit the efficiency of a CRISPR/Cas system. *See, e.g.*, Ex. 1003 ¶¶ 66, 71; Ex. 1061, 200:14–201:11. Moreover, the particular types of chemical modifications disclosed in Pioneer Hi-Bred and recited in the challenged claims had been known and used for decades to stabilize RNA against unwanted degradation in other systems. Ex. 1003 ¶¶ 50–52, 61–70; *see also* Ex. 1061, 214:6–16. Thus, while the art was somewhat unpredictable in December 2014, it was far from a blank slate with a POSA understanding how the different elements of a CRISPR/Cas system are used and function together, including the role of gRNA; the types of chemical modifications that had been successfully used in other systems to reduce RNA degradation, while preserving functionality; and standard techniques for making gRNAs with the modifications disclosed and exemplified in Pioneer Hi-Bred.

Finally, Patent Owner's attempt to analogize the present facts to those in *Impax Laboratories*[14] is unavailing. *See* Resp. 42–43. The claims in *Impax Laboratories* were directed to methods of using a particular

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[14] *Impax Labs., Inc. v. Aventis Pharms., Inc.*, 545 F.3d 1312, 1314 (Fed. Cir. 2008).

compound to treat a particular disease. 545 F.3d at 1314. However, the allegedly anticipating prior art disclosed “hundreds or thousands of compounds and several diseases” along with “broad and general” dosage guidelines and “without sufficient direction or guidance to prescribe a treatment regimen.” *Id.* at 1315–16. On those facts, the Federal Circuit affirmed the district court’s finding that the prior art did not enable the particular method recited in the claims. *Id.* In contrast, Pioneer Hi-Bred exemplifies particular crRNA sequences having the recited chemical modifications at the recited locations and teaches that gRNA comprising such may be used as guide polynucleotides in a CRISPR Cas system.

In sum, given the guidance in Pioneer Hi-Bred, the existing knowledge in the art, including knowledge of standard techniques and equipment/reagents for making the chemically modified RNA sequences taught in Pioneer Hi-Bred, and the relatively high level of training of a POSA, we find that undue experimentation would not have been required to make and use a gRNA with the recited chemical modifications and functionality. Because Pioneer Hi-Bred discloses functional, chemically-modified gRNAs as recited in claims 1 and 19, and that disclosure is enabled, Petitioner has demonstrated that claims 1 and 19 are anticipated by Pioneer Hi-Bred.

ii. Claims 2 and 33

Claim 2 recites a method comprising the steps of: (a) contacting a DNA sequence, gene of interest, or target polynucleotide with a CRISPR-associated (Cas) protein and the gRNA of claim 1, and (b) editing, regulating, cleaving, or binding the DNA sequence, gene of interest, or target polynucleotide. Ex. 1001, 257:48–56. Claim 33 recites the same method using the crRNA of claim 19. *Id.* at 260:20–28.

Petitioner cites evidence showing that Pioneer Hi-Bred discloses the method recited in claims 2 and 33. *See* Pet. 30–32, 47. Patent Owner does not dispute Petitioner's showing for these claims separately from its arguments for claims 1 and 19.[15]

Petitioner's contentions are supported by the record and persuasive. As explained above, Pioneer Hi-Bred discloses a synthetic gRNA and crRNA with the chemical modifications and gRNA functionality recited in claims 1 and 19. Pioneer Hi-Bred discloses methods of using such to modify a DNA sequence. Ex. 1006, 41:23–32, 128:8–14 (claim 28); Ex. 1003 ¶¶ 157–58. Pioneer Hi-Bred also discloses that the gRNA/Cas endonuclease system can be used to regulate expression of a specific gene or cleave a target polynucleotide. *See* Ex. 1006, 30:17–22 (cleave by "introduc[ing] a double strand break at said target site"); 47:15–29 (regulate expression); Ex. 1003 ¶¶ 159–61. Moreover, Dr. Furneaux offers testimony, which we credit, explaining how Pioneer Hi-Bred discloses both the contacting and editing, regulating, cleaving, or binding steps recited in claims 2 and 33. Ex. 1003 ¶¶ 162–66. The teachings Petitioner and Dr. Furneaux rely upon are presumptively enabling. *See, e.g.*, *Antor Media*, 689 F.3d at 1287–88. Patent Owner does not challenge that presumption separately from its arguments for independent claims 1 and 19, which are unavailing as explained above.

[15] To the extent Patent Owner reasserts the same arguments for claims 2, 33, and the other dependent claims that it does for claims 1 and 19, those arguments are unavailing for the reasons already addressed in our analysis above.

Accordingly, Petitioner has shown by a preponderance of the evidence that claims 2 and 33 are anticipated by Pioneer Hi-Bred.

iii. Claim 3

Claim 3 recites "[a] set or a library comprising two or more synthetic guide RNAs of claim 1." Ex. 1001, 257:57–58.

Petitioner relies on Pioneer Hi-Bred's teaching that the "modified guide polynucleotides described [therein] may also be delivered ***simultaneously in multiplex*** to target multiple chromosomal DNA sequences for cleavage or nicking." Pet. 32 (quoting Ex. 1006, 107:24–108:2). In addition, Petitioner cites Pioneer Hi-Bred's teaching that the guide polynucleotide/Cas system can be used "for producing transgenic trait loci ***comprising multiple transgenes***." *Id.* at 50–51 (quoting Ex. 1006, 78:19–26). Petitioner and Dr. Furneaux assert that "[a] POSA would have understood [these teachings] as an express instruction to prepare a library of modified gRNAs comprising multiple modified gRNAs" as recited in claim 30. Pet. 32; Ex. 1003, 166–70.

In response, Patent Owner asserts "[s]equence 66 and 68 do not comprise a library of two or more synthetic guide RNAs, and there [sic] cannot anticipate claim 3." Resp. 28. According to Patent Owner, "Pioneer Hi-Bred does not disclose even one synthetic guide RNA that meets all of the limitations of Claim 1. By necessity, then, it cannot disclose 'two or more' such guide RNAs" as recited in claim 3. *Id.* at 45. Patent Owner does not respond to Petitioner's argument that Pioneer Hi-Bred teaches its modified gRNAs may be delivered in multiplex. *See* Reply 2–3 (noting that

"Patent Owner presents no rebuttal to Petitioner's citation" to this disclosure).

Petitioner's contentions are sufficiently supported by the record and persuasive. As explained above, Pioneer Hi-Bred discloses synthetic gRNA comprising a guide sequence with the recited modifications and gRNA functionality and specifically exemplifies such in sequences 66 and 68 of Table 8. As we understand it, Patent Owner's argument against Petitioner's showing for claim 3 is that a single gRNA comprising either sequence 66 or 68 would not itself be a set or library comprising at least two guide RNAs. But that argument misses the point. Pioneer Hi-Bred specifically teaches that its modified gRNA may be delivered simultaneously in multiplex to target multiple different sequences. Ex. 1006, 107: 24–108:2. Dr. Furneaux offers testimony, which we credit, explaining that a POSA would understand this to be an instruction to prepare a library of multiple, modified gRNAs. Ex. 1003 ¶ 227. Accordingly, Petitioner has shown by a preponderance of the evidence that claim 3 is anticipated by Pioneer Hi-Bred.

iv. Claim 4

Claim 4 depends from claim 1 and recites that the synthetic guide RNA is a "single-guide RNA (sgRNA)." Ex. 1001, 257:60–61. Petitioner asserts that "Pioneer Hi-Bred discloses a single-guide RNA (also known as 'long guide RNA') comprising a VT domain and CER domain" and teaches that "nucleotide modifications, such as 2'-O-methyl and 3-phosphorothioate modifications, e.g., as shown in Table 8, can be made in the guide sequence (VT Domain) of an sgRNA." Pet. 33–34 (citing Ex. 1006, 24:6–19, 25:11–15, 113:25–29, Figs. 1B and 3B; Ex. 1003 ¶¶ 171–76). We agree that the cited evidence shows that Pioneer Hi-Bred discloses a chemically-modified sgRNA as recited in claim 4.

Patent Owner argues that claim 4 is not anticipated because "[s]equences 64, 65, 66, 67, 68, and 69 [in Pioneer Hi-Bred Table 8] are disclosed only as a portion of a duplex or two-part guide." Resp. 29.[16] Patent Owner's argument is unavailing. Pioneer Hi-Bred's disclosure is not limited to a two-part guide with the crRNA sequences in Table 8. It discloses that its guide polynucleotides can be implemented as "a single molecule or a double molecule." 1006, 24:8–9. Moreover, Pioneer Hi-Bred specifically states that the modifications in Table 8 can also be introduced in a "long guide RNA," i.e., a sgRNA. Ex. 1006, 113:25–27; *see* Ex. 1003 ¶ 150 (explaining that a sgRNA is also referred to as a long guide RNA). Accordingly, while sequences 64–69 are described as part of a crRNA, a POSA would have immediately envisioned that those sequences could also be implemented in the corresponding domains of a sgRNA. Thus, Pioneer Hi-Bred anticipates claim 4.

v. Claims 5, 8–13, 20, 21, and 24–28

Claims 5, 8–13, 20, 21, and 24–28 further recite the use of at least one phosphorothioate modification along with the existing 2'-O-methyl modification required by the independent claims. More specifically, claims 5 and 21 recite that the "one or more modifications" in their respective independent claims "comprise[] a 2'-O-methyl nucleotide with a 3'-

[16] To the extent Patent Owner reasserts the same arguments for claim 2 and the other dependent claims that it does for claims 1 and 12, those arguments are unavailing for the reasons already addressed in our analysis above.

phosphorothioate." Ex. 1001, 257:62–64, 259:15–17. Claims 8 and 20 recite that the gRNA and crRNA in their respective independent claims further comprise at least one "phosphorothioate internucleotide linkage, phosphonoacetate (PACE) internucleotide linkage, and/or thiophosphonoacetate (thioPACE) internucleotide linkage." *Id.* at 258:37–40, 259:10–14. Claims 9–11 and 24–26 recite that the gRNA and crRNA in their respective independent claims comprise "up to"[17] a specified number (i.e., 3, 7, or 10) of the same set "internucleotide linkage" modifications "in the guide sequence." *Id.* at 258:41–49, 259:24–260:3. Similarly, claims 12, 13, 27, and 28 recite "up to five consecutive" internucleotide linkage modifications at the 5'-end (claims 12 and 13) and 3'-end (claims 23 and 28). *Id.* at 258:50–57, 260:4–11.

For all these claims, Petitioner points to disclosure in Tables 7 and 8 of Pioneer Hi-Bred, urging that they disclose "five types of nucleotide chemical modifications including the phosphorothioate and 2'-O-methyl modifications." Pet. 34–35 (citing Ex. 1003 ¶ 178–82); *see also id.*at 38 (explaining that sequence 64 in Table 8 discloses "consecutive

[17] The phrase "up to" puts an upper limit on the number of internucleotide linkage modifications. Accordingly, these claims encompass gRNA and crRNA having anywhere from one such modification "up to" the recited number of modifications. Neither party addresses whether the "up to" language puts any lower limit on the required number of internucleotide linkage modifications, i.e., whether the disclosure of a synthetic gRNA or crRNA as recited in the independent claims, but with *none* of the recited internucleotide linkage modifications would anticipate claims 9–13 and 24–28 because zero is less than the recited threshold. We need not resolve this issue because we find that Pioneer Hi-Bred discloses molecules with both 2'-O-methyl and at least one phosphorothioate modification in the guide sequence.

phosphorothioate internucleotide linkages in the gRNA guide sequence (VT domain) at the 5'-end"). Relying on Pioneer Hi-Bred's teaching that these modifications can be introduced in "combination," Petitioner cites Dr. Furneaux's calculations showing that "Tables 7 and 8 disclose thirty-one different ways to combine the five modifications, and eight of these combinations contain both a 3'-phosphorothioate modified nucleotide and a 3'-phosphorothioate modified nucleotide." *Id.* at 35–36 (citing Ex. 1003 ¶¶ 183–185; Ex. 1006, 108:25–27, 113:25–29). Petitioner relies on a similar mathematical analysis based on another teaching in Pioneer Hi-Bred that "yields 2047 ways to combine the [disclosed modifications], 512 of which contain a 2'-O-methyl-3'phosphorothioate modification in the guide sequence." *Id.* at 36 (citing Ex. 1003 ¶ 191; Ex. 1006, 24:6–19). According to Petitioner, a POSA would "immediately envisage combining these modifications in one nucleotide of the guide sequence" given "the high fraction of combinations containing the combined [modifications] and Pioneer Hi-Bred's express teachings of combining" them. *Id.* at 36–37 (citing *Kennametal, Inc. v. Ingersoll Cutting Tool Co.*, 780 F.3d 1376, 1381 (Fed. Cir. 2015)).

Patent Owner argues that sequences 66 and 68 in Table 8 disclose only 2'-O-methyl modifications and therefore cannot anticipate claims 5, 8–13, 20, 21, and 24–28. *See* Resp. 27–28. According to Patent Owner, "Pioneer Hi-Bred simply does not disclose any of the combinations of chemical modifications" required by these claims and "Petitioner admits as much in its analysis" by relying on "Dr. Furneaux's calculation that there are 2047 possible combinations of the 11 specified chemical modifications early in the reference, but notably not in Examples 4 and 5.” *Id.* at 28 (citing Ex. 2025 ¶¶ 255–57).

In its Sur-reply, Patent Owner further asserts that *Kennametal* “is inapposite since [there] it was undisputed that all claim elements were disclosed in the prior art but for a single structural element,” whereas here “it is undisputed that Pioneer Hi-Bred lacks the structural limitations and the ‘gRNA functionality’ limitation.” Sur-reply 26. However, Patent Owner does not challenge Dr. Furneaux’s calculations, nor Petitioner’s argument based on those calculations that a high fraction of the possible combinations would have both modifications. *See* Reply 2 (noting that Patent Owner “presents no analysis or argument” regarding these claims).

Petitioner’s contentions are sufficiently supported by the record and persuasive. Sequence 64 in Table 8 exemplifies phosphorothioate modifications between the three nucleotides at the 5’-end of the guide sequence (VT domain). Ex. 1006, 109. Sequence 66 discloses 2’-O-Methyl modifications at the same location. *Id.* While none of the exemplary sequences in Table 8 disclose the combination of these two modifications to the same nucleotide, Pioneer Hi-Bred discloses that “[n]ucleotide base *and*/or phosphodiester bond modifications similar to those illustrated in Example 5 Table 8 can be introduced individually or *in combination* into the crRNA.” *Id.* at 113:25–27 (emphases added). The combination of the modifications in sequences 64 and 66, for example, would result in a crRNA having 2’-O-methyl-3’-phosphorothioate nucleotides at the 5’-end of the guide sequence.

Regarding claims 13 and 28, sequence 65 in Table 8 exemplifies phosphorothioate modifications between the three nucleotides at the 3’-end of a crRNA. Ex. 1006, 109. The combination of modifications in sequences 65 and 66, for example, would result in a crRNA having 2-O-methyl modifications in its guide sequence and phosphorothioate modifications at the 3'-end of the crRNA. Thus, Pioneer Hi-Bred discloses the recited structural limitations (i.e., 2'-O-methyl and phosphorothioate internucleotide linkage modifications) and expressly teaches that they can be combined in the guide sequence of a guide polynucleotide.[18] For this reason, Patent Owner's attempt to distinguish *Kennametal* is unavailing. *See Kennametal*, 780 F.3d at 1382 (explaining that where a reference discloses all of the claim elements "with the exception of" the recited combination of two of those elements "the question for purposes of anticipation is whether the number of categories and components disclosed in [the reference] is so large that the [recited] combination . . . would not be immediately apparent to one of ordinary skill in the art") (internal quotations omitted).

Here, Petitioner shows that a high fraction of the possible combinations disclosed in Tables 7 and 8 and elsewhere in Pioneer Hi-Bred result in the recited combinations of 2'-O-methyl and 3'-phosphorothioate modifications. Pet. 34–38 (citing Ex. 1003 ¶¶ 171–74). This showing, which Patent Owner does not specifically dispute, sufficiently demonstrates that a POSA would immediately envisage the combination of modifications recited in claims 5, 8–13, 20, 21, and 24–28. Accordingly, Petitioner has shown by a preponderance of the evidence that Pioneer Hi-Bred anticipates these claims.

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[18] As explained above, Pioneer Hi-Bred discloses that its guide polynucleotides have the recited "gRNA functionality." *Supra* § III.E.i.

vi. Claims 14 and 29

Claims 14 and 29 depend from claims 1 and 19 and further recite "a fluorophore at a 5'-end" of the gRNA or crRNA. Ex. 1001, 258:57–58, 60:12–13.

Petitioner contends that "Pioneer Hi-Bred teaches that fluorophore modifications can be combined with 2'-O-methyl modifications in the guide sequence." Pet. 41 (citing Ex. 1006, 27:3–19). According to Petitioner, "the guide sequence (VT domain) is found at either the 5' or 3'-end of the gRNA." *Id.* (citing Ex. 1003 ¶ 211). Thus, "a POSA would immediately envisage gRNAs with a combination of 2'-O-methyl modifications in the guide sequence and fluorophores at the 5'-end." *Id.*

Patent Owner disputes Petitioner's contention. According to Patent Owner, "[t]he only section in Pioneer Hi-Bred that Petitioner identifies" as disclosing this limitation "provides neither an explicit nor inherent disclosure of 'a fluorophore at a 5'-end of the [guide RNA/crRNA].'" Resp. 45 (alterations in original).

Petitioner's contentions are sufficiently supported by the record and persuasive. Pioneer Hi-Bred discloses modifying the VT domain of a guide polynucleotide to provide for "tracking" by adding the benefit of "a fluorescent label." Ex. 1006, 27:3–19. In the same section, Pioneer Hi-Bred teaches that such modifications may be used in "combination" with other modifications to the VT domain such as "a 2'-O-Methyl RNA nucleotide." *Id.* Thus, Pioneer Hi-Bred discloses the combination of 2'-O-methyl and fluorophore modifications in the VT domain of a gRNA and crRNA molecule.

Patent Owner does not clearly explain why, in its view, the disclosure Petitioner cites fails to disclose "a fluorophore at the 5'-end." To the extent Patent Owner contends that it doesn't specify a particular location for the fluorophore, we disagree. Dr. Furneaux offers testimony, which Patent Owner does not specifically dispute, that a POSA would understand that the VT domain is either at the 3' or the 5' end of the molecule. Ex. 1003 ¶ 211. Given that Pioneer Hi-Bred discloses adding a fluorescent label to the VT domain, and that a POSA would know that the VT domain is at one of the two ends of the molecule, Dr. Furneaux testifies "[a] POSA can immediately envisage guide RNAs comprising a 2'-O-methyl modified nucleotide in the guide sequence in combination with a fluorophore at the 5' end." Ex. 1003 ¶¶ 210–11. We find this explanation to be credible and supported by the record. Accordingly, Petitioner has shown by a preponderance of the evidence that claims 14 and 29 are anticipated by Pioneer Hi-Bred.

vii. Claims 15–17, 30 and 31

Claim 15 depends from claim 1 and recites that the gRNA comprises "one or more end modification." Ex. 1001, 258:59–60. Petitioner has shown that Pioneer Hi-Bred discloses a gRNA with an end modification. For example, sequences 66 and 68 in Table 8 disclose 2'-O-methyl modifications of the nucleotides at the 5' end of a crRNA. Ex. 1006, 109. Patent Owner does not dispute Petitioner's showing for claim 15 separately from its arguments for the independent claims, which are unavailing as explained above.

Claims 16 and 30 recite "at least 2 consecutive 2'-O-methyl modifications." Ex. 1001, 258:62–63, 260:14–15. Petitioner again points to sequences 66 and 68 in Table 8, which it contends disclose this limitation. Pet. 42, 46.

Petitioner's contentions are sufficiently supported by the record and persuasive. Sequence 66 discloses a crRNA with three consecutive 2'-O-methyl modifications in the guide sequence. Ex. 1006, 109. Sequence 68 discloses a crRNA with seventeen consecutive 2'-O-methyl modifications in the guide sequence. *Id.*

Claims 17 and 31 recite "at least six 2'-O-methyl modifications." Ex. 1001, 258:64–67, 260:16–17. Sequence 68 discloses this limitation. Thus to the extent Patent Owner contends that Pioneer Hi-Bred does not disclose the "multiple '2'-O-methyl modifications' recited in" claims 16, 17, 30, and 31, it is plainly incorrect. *See* Resp. 46. Accordingly, Petitioner has shown by a preponderance of the evidence that claims 15–17, 30 and 31 are anticipated by Pioneer Hi-Bred.

viii. Claims 18 and 32

Claims 18 and 32 recite "at least twenty 2'-O-methyl modifications." Ex. 1001, 258:66–67, 260:18–19. Petitioner argues that Pioneer Hi-Bred discloses: (1) "making 2'-O-methyl modifications at every position of the guide sequence (VT domain)" and (2) "that the length of the target sequence, and therefore the length of the guide sequence, can be at least twenty nucleotides." Pet. 43–44 (Ex. 1003 ¶¶ 215–18). In support of its argument, Petitioner cites the sequences in Table 8 and further quotes the description of Table 7, which provides that "modifications can be introduced throughout the nucleic acid sequences comprising the VT or CER domains." *Id.* (quoting Ex. 1006:13–25). Based on this disclosure, Petitioner contends "[a]

POSA would immediately envisage utilizing at least twenty 2'-O-methyl modifications when [sic] guide sequences that are at least twenty nucleotides in length." *Id.* at 44 (citing Ex. 1003 ¶ 218).

Patent Owner disputes Petitioner's contentions, explaining that sequences 66 and 68 disclose three and seventeen 2'-O-methyl modifications, but not the "at least twenty" such modifications recited in these claims. Resp. 46. According to Patent Owner, Petitioner's argument is flawed because "modification 'throughout the nucleic acid sequences' does not mean more than twenty modifications. Rather the phrase could just as easily mean less than twenty modifications interspersed throughout the entire length of the polynucleotide." *Id.* (referring to Petitioner's reliance on the disclosure at Ex. 1006, 106:13–25).

Petitioner's contentions are sufficiently supported by the record and persuasive. Sequence 68 in Table 8 discloses 2'-O-methyl modifications to every nucleotide in the guide sequence (i.e., the VT domain). Ex. 1006, 109. That disclosure supports Petitioner's reading of the phrase "modifications . . . introduced throughout the nucleic acid sequence comprising the VT domain" to mean that every nucleotide in the VT domain is modified. Pet. 43 (quoting Ex. 1006, 106:13–25). The particular guide sequence exemplified in sequence 68 is only 17 nt long, but Pioneer Hi-Bred teaches that the VT domain may vary in length from 12 to 30 nt. Ex. 1006, 3:19–20, 98:19–23. Thus, the record supports Petitioner's argument that a POSA would understand Pioneer Hi-Bred to also disclose gRNAs having a guide sequence of at least twenty nucleotides and we credit Dr. Furneaux's testimony that a POSA would immediately envisage gRNAs with guide sequences of at least 20 nt in which every nucleotide has a 2'-O-methyl modification like exemplary sequence 68. Ex. 1003 ¶ 216–18. Accordingly, Petitioner has shown by a preponderance of the evidence that claims 18 and 32 are anticipated by Pioneer Hi-Bred.

F. Ground 2: Obviousness over Pioneer Hi-Bred and Krutzfeldt, Deleavey, Soutscheck, or Yoo

Petitioner contends that claims 5, 8–13, 18, 20, 21, 24–28, and 32 are obvious over Pioneer Hi-Bred in combination with any of Krutzfeldt, Deleavey, Soutschek, or Yoo. Pet. 47–67.

For claims 5, 8–13, 20, 21, and 24–28, Ground 2 relies on Pioneer Hi-Bred for the limitations of the independent claims and any one of the other references for their disclosure of modified nucleotides comprising a 2'-O-methyl nucleotide with one or more phosphorothioate modifications. Petitioner asserts that "[a] POSA would have been motivated to incorporate the 2'-O-methyl-3'phophorothioate modifications disclosed in Krutzfeldt, Deleavey, Soutschek, or Yoo's RNA molecules with the modified gRNAs taught in Pioneer Hi-Bred for multiple reasons." Pet. 52. Petitioner's reasoning includes that it was "well known to use such modified nucleotides in the targeting sequences of several types of RNA molecules" and the 2'-O-methyl-3'-phosphorothioate modifications in these secondary references "reflect the types of modifications to gRNA that are already taught and suggested in Pioneer Hi-Bred." *Id.* at 52–53. Petitioner further contends that a POSA would have been motivated to make the combination because the references teach that such modifications provide benefits such as improved stability and resistance towards nuclease degradation and increased editing efficiency. *Id.* at 54–56.

According to Petitioner, a POSA would have had a reasonable expectation of success because "it is undisputed that gRNAs with 2'-O-methyl and phosphorothioate modifications in the guide sequences are inherently functional for this purpose." Pet. 57. Petitioner also relies on Pioneer Hi-Bred's teaching that both types of modifications can be used in a gRNA. *Id.* at 57–58. Finally, Petitioner urges that the "use of 2'-O-methyl-3'-phosphorothioate modifications in the targeting sequences of various types of RNAs, such as siRNA, AONs, and anti-miRNA, in the field of RNA therapeutics" was "widespread" and therefore a POSA would have anticipated that the same modifications "would not only have been functional, but also would have had . . . additional benefits" such as increasing stability and binding specificity. *Id.* at 58–59 (citing Ex. 1003 ¶¶ 267–79).

For claims 18 and 32, Ground 2 relies on same disclosure in Pioneer Hi-Bred noted above in Ground 1 and further cites Krutzfeldt as evidence that it was "known in other gene regulation applications that the entire RNA oligonucleotide can be comprised of 2'-O-methyl nucleotides" to support its argument that a gRNA with "at least twenty 2'-O-methyl nucleotides" would have been obvious. *Id.* at 64 (citing Ex. 1003 ¶ 299; Ex. 1009, 2886). Petitioner further asserts that "it was known that the guide sequence for Cas9 gRNAs is commonly 20 nucleotides long." *Id.* at 64–65 (citing Ex. 1013, 827). According to Petitioner,

> Given Pioneer Hi-Bred's express teachings to make gRNAs with guide sequences entirely composed of 2'-O-methyl nucleotides and to target sequences at least twenty nucleotides long, a POSA would have been motivated to make a gRNA with twenty 2'-O-methyl modifications in the guide sequence. A POSA would understand that using the same number of 2'-

> O-methyl modifications as the number of nucleotides in the target site produces recognizable and measurable effects in gene regulatory activity. Therefore, the number of 2'-O-methyl modifications, is a result-effective variable which, when changed, achieves a recognized result.

Id. at 65 (citing Ex. 1003 ¶¶ 301–04).

Petitioner's contentions are sufficiently supported by the record and persuasive. Beginning with claims 5, 8–13, 20, 21, and 24–28, Krutzfeldt, Deleavey, Soutschek, and Yoo each disclose chemically modified RNA sequences having both 2'-O-methyl and phosphorothioate modifications, including those comprising a "2'-O-methyl nucleotide with a 3'-phosphorothioate nucleotide" as recited in claim 5. Ex. 1009, 2886 (Table 1), 2889 (Fig. 5A); Ex. 1007, 943 (Fig. 4), 948; Ex. 1012, 173, 177; Ex. 1011, 2008. Moreover, as Petitioner points out, these references teach that such modifications, particularly when made to nucleotides near the end of an RNA molecule, provide a number of benefits including increased resistance to nuclease degradation. Ex. 1009, 2889; Ex. 1007, 937; Ex. 1012, 173; Ex. 1011, 2008. This dovetails with Pioneer Hi-Bred's teaching that such modifications decrease unwanted nuclease degradation in a guide polynucleotide. *See* Ex. 1006, 107. Accordingly, the combination of Pioneer Hi-Bred with any of Krutzfeldt, Deleavey, Soutschek, or Yoo teaches all of the limitations of these claims and the record supports Petitioner's rationale for combining them.

Regarding claims 18 and 32, we agree with Petitioner that Krutzfeldt teaches anti-miRNAs having 20 or more nucleotides wherein every nucleotide in the sequence has a 2'-O-methyl modification. Ex. 1009, 2886 (Table 1 using lower case letters to indicate 2'-O-methyl-modified nucleotides). Krutzfeldt refers to these as "antagomirs," which it describes as "RNA-like oligonucleotides that harbor various modifications for RNAse protection and pharmacologic properties such as enhanced tissue and cellular uptake." *Id.* at 2885; *see also id.* at 2889 (teaching these modifications protect against exonucleases and endonucleases). Moreover, Krutzfeldt teaches that antagomirs having 2'-O-methyl modifications at every nucleotide and numerous phosphorothioate modifications can still form a duplex with the complementary miRNA. Ex. 1009, 2890-91; *see also* Ex. 1007, 948 (citing Krutzfeldt as showing that such antagomirs are "a successful strategy for targeting miRNAs"). Dr. Furneaux offers testimony, which we credit, that this fact supports that a POSA would have reasonably expected "that a guide RNA with the same modification in its targeting sequence (i.e., the guide sequence) would bind to a complementary region of a polynucleotide in vivo." *Id.* ¶¶ 274-77.

Accordingly, the disclosure in Krutzfeldt adds to the existing teachings in Pioneer Hi-Bred that 2'-O-methyl modifications can be made to every nucleotide in the guide sequence. Ex. 1006, 109 (Table 8 sequence 68), 106:13–25. Given that Pioneer Hi-Bred teaches that the length of the guide sequence is variable and can be 20 or more nt long depending on the target, we agree that these teachings support Petitioner's obviousness argument. Ex. 1006, 32:1–3; *see also id.* at 3:19–20, 98:19–23. That is, even if a POSA would not have immediately envisaged a gRNA having at least 20 2'-O-methyl modifications in its guide sequence from the disclosure in Pioneer Hi-Bred, they would have considered such a gRNA to be the obvious result of routine optimization to a slightly longer guide sequence having at least 20 nt. Ex. 1003 ¶¶ 301–03.

Patent Owner raises several arguments in its Response some of which were already addressed in our analysis of Ground 1. For example, Patent Owner urges that all of Petitioner's obviousness grounds fail because they rely exclusively on Pioneer Hi-Bred for the claimed functionality requirements. *See* Resp. 50–51. However, as explained above, Pioneer Hi-Bred discloses synthetic gRNA and crRNA molecules having the recited "gRNA functionality" and the anticipating disclosure is enabled as to both Patent Owner's composition and method claims. Accordingly, Patent Owner's functionality arguments are unavailing.

Patent Owner also challenges Petitioner's motivation to combine and reasonable expectation of success showing and offers evidence of objective indicia of non-obviousness. *See* Resp. 47–50, 51–63. We address these issues in turn.[19]

1. <u>Whether there would have been a motivation to combine and reasonable expectation of success</u>

Patent Owner argues that Petitioner "relies on the claims as a roadmap" and "regardless of the number of or variety of modifications in a challenged claim" the alleged motivation to combine is always the same, i.e., that Pioneer Hi-Bred envisions other combinations and "the modifications in the prior art reference would enhance the protection against degradation." Resp. 51–52. According to Patent Owner, "protecting against degradation is a function the disclosed modifications in Pioneer Hi-Bred were already

[19] These are global arguments Patent Owner collectively argues for all of the obviousness grounds. Our analysis below considers the claims in all of those grounds.

providing so a POSA would have no need or motivation to look elsewhere to an already fulfilled need." *Id.* at 52.

Patent Owner's argument is unavailing for several reasons. First, at least with respect to claims 5, 8–13, 18, 20, 21, 24–28, and 32 Pioneer Hi-Bred teaches the same modifications (i.e., 2'-O-methyl sugar and phosphorothioate linkage modifications) taught by the secondary references in Ground 2. Thus, the choice between Pioneer Hi-Bred's modifications and those taught in the Krutzfeldt, Delevey, Soutschek, and Yoo that Patent Owner's argument envisions is an illusory one.

Second, even if the combination involved the use of modifications not already taught in Pioneer Hi-Bred,[20] it is well-established that substitution of one element for another known to provide the same benefit may provide a rationale for combining references. *See KSR*, 550 U.S. at 417 ("[I]f a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill."). So too, "the normal desire to improve upon what is already generally known" may provide a reason to employ additional types of modifications as in claims 6, 7, 22, and 23; to combine modifications as in claims 5, 8–13, 20, 21, and 24–28; or to use more of the same modifications as in claims 18 and 32, where such strategies had been

[20] This is the case for the PACE and thioPACE modifications taught by Threlfall and Delevey in Ground 3. Pioneer Hi-Bred does not teach those modifications, but Petitioner asserts that it would have been obvious to use them in Pioneer Hi-Bred's synthetic gRNAs because "they would provide the same benefits to gRNAs that Pioneer Hi-Bred seeks to achieve." Pet. 70.

shown to prevent degradation in prior art RNA systems. *Jazz Pharms., Inc. v. Amneal Pharms.,* LLC, 895 F.3d 1347, 1368 (Fed. Cir. 2018).

Third, along with protecting against degradation, Petitioner points to teachings in these references that their chemical modifications provide other benefits, e.g., increased specificity, cell penetration or tracking, that would have motivated a POSA to employ such modifications in Pioneer Hi-Bred's synthetic gRNAs. *See* Pet. 51 (referring to Yoo's teaching regarding "reduced non-specific effects"); 69–70 (referring to Threlfall's teaching regarding "greater efficacy and ability to penetrate cells" and "increased cellular uptake"); 85–86 (citing references showing that "attaching fluorophores to biological molecules . . . was well-known and commonly done, *e.g.*, for purposes of tracking"); Ex. 1003 ¶¶ 267–72 (Ground 2); 323–25 (Ground 3); 391–93 (Ground 6). Moreover, Dr. Furneaux provides testimony explaining why POSA would understand that the various benefits these modifications provide to be desirable within the context of a CRISPR/Cas system. Ex. 1003 ¶¶ 261–62 (Ground 2); 324–25 (Ground 3); 394 (Ground 6). This testimony is credible and further supports Petitioner's reasoning for its obviousness combinations.

Patent Owner also argues there would have been no reasonable expectation of success in making the chemical modifications in Petitioner's obviousness combinations. *See* Resp. 52–63. According to Patent Owner, "[w]hether a particular combination of chemical modifications in at [sic] particular nucleotide positions on the give [sic] would adversely impact functionality was entirely unpredictable at the time Agilent started its work, and there was no guarantee of success that it would make the discoveries that led to the claimed inventions." *Id.* at 52–53. Patent Owner explains that

"CRISPR was a nascent technology in 2014, and the structure-function relationships among the gRNA and Cas protein were still being investigated." *Id.* at 54–59 (citing references). Therefore, urges Patent Owner, "a POSA would know that the result of modifications would be unpredictable and would need to be tested." *Id.* at 55.

Regarding Petitioner's reliance on modifications known to be successful in prior art RNA systems, Patent Owner contends that a POSA would not have regarded such systems as "'mechanistically analogous' such that modifications that maintained functionality in other systems would work in the CRISPR-Cas system." Resp. 59–60 (citing Ex. 2025 ¶¶ 132–47). Patent Owner argues that Dr. Furneaux's testimony that "prior art siRNA, miRNA, AON" systems are "mechanistically analogous" to CRISPR/Cas systems is flawed because "notably absent from his [declaration] ***is any description of the mechanisms of the systems that are allegedly 'mechanistically analogous.'***" *Id.* at 61–62 (citing Ex. 1003 ¶¶ 43–49, 61–78). Patent Owner urges that Petitioner and Dr. Furneaux's "entire 'mechanistically analogous' analysis reduces simply to noting that RNA, which is common to all these systems, faces the same problems, such as potential degradation." *Id.* at 62.

Finally, Patent Owner argues that "Agilent's own process confirms that arriving at the claimed combinations was anything but predictable or that a POSA would have had a reasonable likelihood of success." Resp. 63 (referring to Dr. Sampson's testimony). According to Patent Owner, this was "an iterative process as to which there was no assurance of success via which Agilent was able to test its various modified guides across a series of assays sufficient to appraise the public on how to improve CRISPR gRNAs." *Id.*

In reply, Petitioner argues that Patent Owner's arguments and Dr. Marshall's testimony regarding the complexity of the CRISPR/Cas system and mechanistic differences with prior art systems are "theoretical concerns . . . that were never actually expressed in the literature." Reply 23–25. According to Petitioner, by asserting that these "speculative and contrived concerns can defeat obviousness, Patent Owner effectively applies a heightened standard that goes beyond reasonable expectation of success." *Id.* at 25. Instead, "the law is clear that 'the expectation of success need only be reasonable, not absolute.'" *Id.* (quoting *Pfizer, Inc. v. Apotex, Inc.*, 480 F.3d 1348, 1364 (Fed. Cir. 2007)).

Petitioner urges that the record supports such a finding here because;

> [r]esearchers had been using the claimed chemical modifications in other relevant gene regulation contexts long before the conception of CRISPR as a gene editing tool. Then, when Doudna and co-workers published CRISPR for gene editing,[21] multiple researchers ***immediately*** proposed to use such chemical modifications with gRNAs, including the specific claimed 2'-O-methyl and phosphorothioate modifications.

Reply 26 (citing Pet. 12, 81; Ex. 1003 ¶¶ 71–72; Ex. 1061, 112:1–5, 113:3–8, 116:5–10, 211:14–212:8). According to Petitioner, "[i]f there were no reasonable expectation of success, it would not be the case that a chorus of

[21] "In 2012, Jennifer Doudna and Emmanuelle Charpentier discovered [and published] that CRISPR associated enzyme (more specifically, the Cas9 protein) could be programmed to target nearly any portion of a genome using lab-made guide RNA sequences." Ex. 2025 ¶ 77. Their work was published in June 2012. Resp. 53; Reply 20.

researchers would so quickly propose using chemical modifications with gRNAs while none would warn against it." *Id.* at 26. Petitioner also points to evidence that prior art "studies ha[d] shown that modifications at the 5' or 3'-ends of gRNAs do not inhibit Cas9 function" and "crystal structure analysis of Cas9 show[ed] that it tolerates a large number of mutations to the gRNA, including multiple modifications to single nucleotides." *Id.* at 27 (citing Ex. 1003 ¶¶ 78; Ex. 1026, 941; Ex. 1033, 279; Ex. 1034, 235).

Petitioner's arguments and evidence are persuasive. As Petitioner points out, by December 2014, several studies had shown that the CRISPR/Cas system could successfully tolerate modifications. Ex. 1026, 941; Ex. 1033, 279. While these studies describe different types of modifications[22] than those in the challenged claims, such evidence nevertheless supports Dr. Furneaux's testimony that a POSA would have expected that the recited chemical modifications could be made to a gRNA while preserving the Cas enzyme's gene editing function. Ex. 1003 ¶¶ 78, 273–79, 301-03, 326-28, 395.

The record further demonstrates that shortly after the discovery of the CRISPR/Cas system for gene editing and prior to December 2014, there were already a number of researchers in addition to the authors of the Pioneer Hi-Bred publication suggesting the use of the claimed chemical modifications to improve the resistance of gRNA to degradation. Ex. 1003 ¶ 71 (identifying examples of patent publications with filing dates prior to

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[22] As Patent Owner points out, the "mutations" referred to in Exhibit 1026 are changes to the RNA bases, as opposed to modifications to the sugar or phosphodiester linkages in the backbone. Sur-reply 12 (citing Ex. 1026, 942–43 (Fig. 4D)). Exhibit 1033 describes truncation, i.e., removal of nucleotides, within the gRNA.

December 2014 describing 2'-O-methyl and phosphorothioate modifications to gRNA); Ex. 1019 ¶¶ 193–96, claim 171; Ex. 1020 ¶¶ 570–71, 1657; Ex. 1022 ¶ 260; Ex. 1023 ¶ 362. Patent Owner's expert, Dr. Marshall, conceded as much on cross-examination. Ex. 1061, 113:3–8, 116:5–10, 211:14–213:8. The fact that multiple groups of researchers independently suggested the same types of gRNA modifications recited in the challenged claims evidences that a POSA would have had a reasonable expectation those modifications could be successfully employed in a CRISPR/Cas system. Ex. 1059 ¶ 18; *see also Regents of the Univ. of California v. Broad Institute, Inc.*, 903 F.3d 1286, 1295 (Fed. Cir. 2018) (explaining that "simultaneous invention" may bear on the obviousness analysis because "it is evidence of the level of skill in the art"). Moreover, while Petitioner points to multiple references suggesting such modifications to gRNA, neither Patent Owner nor Dr. Marshall identify any reference expressing doubt that such modifications could be successfully implemented in a CRISPR/Cas system. This contrast undermines Patent Owner's argument that a POSA would not have reasonably expected the prior art modifications to work in a CRISPR/Cas system.

Nevertheless, Patent Owner maintains "there was no guarantee of success" and therefore testing was required to confirm that such modifications would work. *See* Resp. 51–52. The problem with that argument is that "only a reasonable expectation of success, not a guarantee, is needed" to show obviousness. *Pfizer*, 480 F.3d at 1364 (citing *In re O'Farrell*, 853 F.2d 894, 903 (Fed. Cir. 1988)). The facts of the *Pfizer* case are instructive. There, the lower court had concluded that there would have been no expectation of success in making the claimed drug salt "because there was no reliable way to predict the influence of a particular salt species on the active part of the compound." *Id.* The Federal Circuit reversed, explaining that while it accepted the lower court's finding that "it was generally unpredictable as to whether a particular salt would form and what its exact properties would be," the "case law is clear that obviousness cannot be avoided simply by a showing of some degree of unpredictability in the art." The Federal Circuit observed that

> a rule of law equating unpredictability to patentability, applied in this case, would mean that any new salt—including those specifically listed in the [prior art reference]—would be separately patentable, simply because the formation and properties of each salt must be verified through testing. This cannot be the proper standard since the expectation of success need only be reasonable, not absolute.

Id.

The current proceeding presents analogous facts. We do not doubt that there would have been some degree of unpredictability because, as Patent Owner points out, the interactions in a CRISPR/Cas system are complex and that there are mechanistic differences from the prior art RNA systems in Petitioner's secondary references. *See* Resp. 53–63. However, Petitioner has shown that a POSA would have understood that these systems share features in common with CRISPR/Cas, including that their efficacy and functionality is limited by the fact that the RNA used in them is subject to unwanted degradation. *See, e.g.*, Ex. 1003 ¶¶ 65–66, 70–72, 261–62, 323–25. Thus, the fact that the claimed modifications reduced degradation, while maintaining functionality, in these prior art systems supports a finding that the same modifications could also be successful in a CRISPR/Cas system. To the extent Dr. Marshall testifies that the successful use of these modifications in prior art systems does not evidence at least a reasonable expectation they could also be successfully used in a CRISPR/Cas system (*see, e.g.*, Ex. 2025 ¶¶ 132, 147, 157), we credit Dr. Furneaux's testimony (Ex. 1003 ¶¶ 61–80, 263–79, 303 (Ground 2), 326–28 (Ground 3), 395 (Ground 6)) and the evidence noted above over Dr. Marshall's testimony on this issue.

For these reasons, Petitioner has sufficiently shown that a POSA would have had both a motivation for and reasonable expectation of success for its obviousness combinations.

2. Consideration of objective indicia of nonobviousness

Patent Owner argues that "[t]here is overwhelming evidence of secondary consideration of nonobviousness, particularly as it relates to the question about [sic] industry praise, copying, and commercial success." Resp. 47–48. Patent Owner's objective indicia arguments center on the Hendel paper (Ex. 1005). *See id.* at 48–50. Patent Owner explains that "[t]he Agilent inventions" were "first made public" in this paper, which was co-authored with researchers at Stanford. *Id.* at 19. According to Patent Owner, "publications citing the Hendel paper have called Agilent's work 'pioneering,' 'seminal,' and 'a major contribution.'" *Id.* at 48 (quoting Ex. 2050, 4; Ex. 2028, 681). Patent Owner also offers evidence that the Hendel paper has been cited almost 900 times since its publication in 2015, which is in the 94th, 98th, or 99th percentile according to various indices of "tracked articles of a similar age." *Id.* at 20, 47 (citing Ex. 2025 ¶ 57; Ex. 2032; Ex. 2056).

Regarding copying and commercial success, Patent Owner points to evidence of statements by Petitioner that it contends "tout the benefits of

Agilent's inventions" and suggest that the reason Petitioner uses chemically modified gRNA in its products is the study published in the Hendel paper. Resp. 20, 48–49 (quoting statements by Synthego's Head of Synthetic Biology in Ex. 2033, 3:05–4:45[23] and citing Ex. 2034, 6). Patent Owner also asserts that "[t]here is . . . no doubt that Petitioner Synthego has been successful" as result of using Patent Owner's inventions. *Id.* at 50.

In its Reply, Petitioner urges that the fact that multiple researchers "***immediately*** proposed to use chemical modifications with gRNAs" after the initial publication of CRISPR for gene-editing undermines Patent Owner's objective indicia arguments. *See* Reply 20–22. Moreover, Petitioner asserts that the evidence of praise for the Hendel paper is "surprisingly thin" with one of the two references Patent Owner cites praising it as "'pioneering'" having been "co-authored by Hendel himself." *Id.* at 21 (referring to Ex. 2050). According to Petitioner,

> Patent Owner's scientists may well have been the first to synthesize some chemically modified gRNAs and do some testing experiments, and they may have been the first to report such work in an article that has been frequently cited over the years. But Patent Owner points to no evidence that this frequent citation stems from a belief among scientists that the authors had done something inventive as opposed to merely generating confirmatory data.

Id. at 22.

Petitioner further contends that it did not copy the Hendel paper, but "simply uses the same tried and true techniques that had long been known in

[23] Exhibit 2033 is a video. The pinpoint citation refers to the time in the video at which these statements occur, i.e., beginning at 3 minutes and 5 seconds.

the art." According to Petitioner, the claimed chemical modifications and the idea to use them in gRNA were already in the prior art therefore "such modifications lack the required nexus for secondary considerations." Reply 23 (citing *In re Huai-Hung Kao*, 639 F.3d 1057, 1068 (Fed. Cir. 2011)). Finally, Petitioner asserts that the evidence of objective indicia of nonobviousness "simply cannot overcome" the strength of the other evidence demonstrating obviousness. *Id.* (quoting *Wyers v. Master Lock Co.*, 616 F.3d 1231, 1246 (Fed. Cir. 2010)).

In its Sur-reply, Patent Owner argues that the data in the Specification presents unexpected results. Sur-reply 21–22. Patent Owner further contends that Petitioner and its experts "suffer from gross credibility problems, ignoring Synthego's admissions of non-obviousness" and "contradicting Synthego's own praise for the Hendel paper" as a "'landmark'" publication. *Id.* at 22–23 (quoting Ex. 2033, 3:05–4:45).

We assess the parties' objective indicia arguments below and then weigh any evidence of such with the other evidence of record to reach a conclusion on Petitioner's obviousness grounds.

Beginning with commercial success, we find Patent Owner's evidence and arguments unavailing. First, Patent Owner cites no evidence to support its factual assertions regarding Synthego's success. *See* Resp. 50 (asserting without citation to evidence of record that Synthego "has raised almost $500 million in funding and recently announced the opening of a 20,000 square foot GMP facility to add to its capacity."). Nor has Patent Owner identified sales or profits stemming either from its own products or any allegedly infringing products to support its allegations of commercial success. Second, even if there was evidence of commercial success, we agree with Petitioner that Patent Owner has not shown that such success is attributable to its use of the claimed chemical modifications as opposed to other factors. *See* Reply 23; *see also* Ex. 1059 ¶ 81 (identifying other business factors to which Petitioner contributes its market performance). Accordingly, Patent Owner's commercial success arguments carry no weight.

Patent Owner's industry praise and copying arguments fare somewhat better. The record supports that the Hendel paper has been heavily cited and that Petitioner's own head of synthetic biology referred to it as a "landmark paper" in a video presentation. Ex. 2025 ¶ 57; Ex. 2033, 3:05–4:45; *see also* Ex. 2034, 6 (stating that the Hendel paper "study set the bar for chemically modified guide RNAs as the method of choice for CRISPR-Cas9 in primary human immune cells"). In the same presentation, Petitioner's executive states that the study in the Hendel paper is the reason Petitioner uses "single guide RNAs in a chemically modified format." Ex. 2033, 3:05–4:45. As such, we find that Patent Owner has presented some evidence of industry praise and copying related to the Hendel paper.

There are, however, significant questions regarding whether there is a nexus between the Hendel paper and any novel aspects of the challenged claims that limit the probative value of this evidence. "Before secondary considerations can carry the day" the proponent of that evidence must establish a nexus with the patent claims at issue. *Huai-Hung Kao*, 639 F.3d at 1068. "Where the secondary consideration actually results from something other than what is both claimed and *novel* in the claim, there is no nexus." *Id.* In this case, the use of 2'-O-methyl and phosphorothioate modifications, as well as other types of modifications to the sugar and internucleotide linkage, in a gRNA was already taught in the prior art.

Ex. 1003 ¶ 71 (citing earlier-filed applications from other researchers suggesting the use of such modifications in gRNA), ¶ 394 (citing earlier and contemporaneously-filed applications from other researchers suggesting the use of a fluorophore at the 5' end of a gRNA); Ex. 1006, 27:3–18, 107–113. This suggests that any praise or copying of the Hendel paper may actually result from its use of these prior art ideas as opposed to any novel aspect of the challenged claims.

In addition, the Hendel paper describes tests of "[c]hemical modifications comprising 2'-O-methyl (M), 2'-O-methyl 3'phosphorothioate (MS), or 2'-O-methyl 3'thioPACE (MSP) . . . at three terminal nucleotides at both the 5' and 3' ends." Ex. 1005, 1. The MS modifications correspond to the modifications recited in the claims in Ground 2 and the MSP modifications are within the scope of the modifications recited in the claims in Ground 3. But it is unclear whether Petitioner's evidence of praise for the Hendel paper results from the MS and MSP modifications in those claims as opposed to other aspects of that study, e.g., the use of M modifications alone.[24] Similarly, Patent Owner's copying evidence suggests that Petitioner uses chemically modified gRNAs, but does not show that those chemically modified gRNAs have the particular modifications recited in the Ground 2, Ground 3, or Ground 6 claims as opposed to other modifications. For these

[24] According to Patent Owner, out of the 895 citations to the Hendel paper only a fraction of those include terms relating to the particular modifications in recited in the claims challenged in Grounds 2 and 3. Resp. 47 (noting that only 276 of the citing papers include the term "phosphorothioate" and only 67 include the term "thioPACE"). This diminishes the argument that there is a nexus between the particular modifications in those dependent claims and the praise that may be inferred from the fact that numerous authors have cited the Hendel paper.

reasons, we give some weight to the evidence of industry praise and copying, however that weight is diminished by the tenuous nexus to the challenged claims.

Moreover, the fact that multiple research groups, nearly simultaneously proposed the use of chemically modified gRNA is itself objective evidence that the challenged claims would have been obvious. *See Regents*, 903 F.3d at 1295 (explaining that "simultaneous invention" is "objective evidence that person of ordinary skill in the art understood the problem and a solution to that problem"); *Geo. M. Martin Co. v. All. Mach. Sys. Int'l LLC*, 618 F.3d 1294, 1305–6 (Fed. Cir. 2010) ("Independently made, simultaneous inventions made within a comparatively short space of time are persuasive evidence that a claimed apparatus was the product of ordinary mechanical or engineering skill.") (internal quotations omitted). As explained above, we credit Petitioner's showing that a POSA would have reasonably expected that such modifications could be successfully made to the gRNA in a CRISPR/Cas system. Thus, the fact that multiple researchers simultaneously proposed making 2'-O-methyl, phosphorothioate, fluorophore, and other modifications to gRNA close in time to the initial publication describing the CRISPR/Cas system for gene editing suggests that this was an obvious solution to a known problem.

Finally, the unexpected results arguments in Patent Owner's Sur-reply are unavailing. Sur-reply 21–22. As an initial matter, these arguments were not presented in the Response such that Petitioner could address them in the Reply. For this reason, Patent Owner's unexpected results argument is untimely and has been waived. Even so, we disagree with Patent Owner's assertion that Petitioner "did not rebut the Patent's explicit discussion of the 'surprising' results obtained or why they were unpredictable." *Id.* (citing Ex. 1001, 3:34–36, 66:27–32, 66:61–67:6). It is Patent Owner as the proponent of this evidence who has the burden to show a nexus between its objective indicia evidence and the merits of the claimed invention. *Kao*, 639 F.3d 1057, 1068; *see also In re* Klosak, 455 F.2d 1077, 1088 (CCPA 1972) ("[T]he burden of showing unexpected results rests on he who asserts them."). The threadbare assertion of allegedly unexpected results in the Sur-reply does not do so. Moreover, Petitioner points out that the results in the Specification show that almost all of the chemically modified gRNAs Patent Owner tested exhibited cleavage activity. Reply 13–14, 17, Ex. 1059 ¶ 47 ("97% of the experiments that Agilent ran showed at least some cleavage activity"). This would seem to be exactly the result a POSA would expect in view of the teachings in Pioneer Hi-Bred and the other references cited in the Petition. For these reasons, Patent Owner's unexpected results arguments carry no weight.

3. Conclusion for Ground 2

Considering the totality of the evidence regarding claims 5, 8–13, 18, 20, 21, 24–28, and 32, including objective indicia of non-obviousness, we determine that Petitioner has established, by a preponderance of the evidence, that these claims would have been obvious over the combination of Pioneer Hi-Bred with any of Krutzfeldt, Deleavey, Soutschek, or Yoo. Indeed, even if Patent Owner had established a sufficient nexus between these claims and its industry praise and copying evidence such that it was entitled to more weight, we would reach the same conclusion given the relative strength of Petitioner's showing. *See, e.g.*, *Bayer Pharma AG v.*

Watson Labs., Inc., 874 F.3d 1316, 1329 (Fed. Cir. 2017) (weighing evidence of unexpected results and copying together with other evidence, including "strong evidence of a motivation to make the claimed combination" in the cited prior art, to conclude that combination was obvious); *Wyers v. Master Lock Co.*, 616 F.3d 1231, 1247 (Fed. Cir. 2010) (listing cases where objective indicia evidence did not overcome a strong case of obviousness).

G. Ground 3: Obviousness over Pioneer Hi-Bred and Threlfall or Deleavey

Petitioner contends claims 6, 7, 22, and 23 are obvious over Pioneer Hi-Bred in combination with either Threlfall or Deleavey. Pet. 67–73. More specifically, Ground 3 relies on Pioneer Hi-Bred for the limitations of the independent claims and Threlfall and Deleavey for their disclosure of "phosphonoacetate" and "phosphonothioacetate" (i.e., PACE and thioPACE) modifications as recited in claims 8 and 16 and "2'-O-methyl-3'-phosphonoacetate" and "2'-O-methyl-3'-phosphonothioacetate" nucleotides as recited in claims 6, 7, 22, and 23. *Id.*

Petitioner asserts that a POSA would have been motivated to use the PACE and thioPACE modifications taught in Threlfall and Deleavey in Pioneer Hi-Bred's gRNA and crRNA molecules "because they would provide the same benefits to gRNAs that Pioneer Hi-Bred seeks to achieve," including "[r]esistance to cellular degradation and increased cellular permeability" and "increased binding affinity." Pet. 70–73 (citing Ex. 1003 ¶¶ 324, 330–33). In addition, Petitioner contends that "Threlfall reports successful cellular uptake of RNAs" with these modifications and therefore "a POSA would have understood that combining and synthesizing gRNAs with" these modifications in the guide sequences of gRNAs "would result in similarly increased cellular uptake of these modified gRNAs." *Id.* at 70; *see also id.* at 73 (urging that a POSA would have understood that PACE and thioPACE "modifications in the targeting sequence would have had similar effects as noted in Threlfall and Deleavey").

According to Petitioner, a POSA would have had a reasonable expectation of success because "RNAs including such modifications had been previously synthesized in Threlfall" and "such synthesis methods were commercially available." Pet. 71 (citing Ex. 1003 ¶ 326). Moreover, Petitioner contends that because of the "previous widespread use" of such modifications "in various types of RNA molecules in the field of gene regulation, a POSA would have reasonably expected" gRNAs containing such modifications "to be functional with the gRNA/Cas system." *Id.* 71–72 (citing Ex. 1003 ¶ 289).

Petitioner's contentions are sufficiently supported by the record and persuasive. As explained above, Pioneer Hi-Bred discloses all of the limitations of the independent and other claims from which claims 6, 7, 22, and 23 depend. Regarding the additional limitations in claims 6, 7, 22, and 23, Threlfall and Deleavey teach that PACE and thioPACE modifications to RNA enhance resistance to degradation and are tolerated in prior art systems. Ex. 1007, 8; Ex. 1010, 1–2. Threlfall further teaches the combination of such modifications with 2'-O-methyl modified nucleotides at the 3' and 5' ends of an RNA molecule. Ex. 1010, 2 (Table 1). Thus, the combination of references articulated in the Petition teaches or suggests all of the limitations of claims 6, 7, 22, and 23.

Moreover, the record supports Petitioner's rationale for combining the references. The teachings in Threlfall and Deleavey support Dr. Furneaux's testimony (Ex. 1003 ¶¶ 320–25, 332–34) regarding the benefits, e.g., increased resistance to degradation and enhanced cellular uptake, that would have motivated a POSA to use such modifications in Pioneer Hi-Bred's gRNA. *See, e.g.*, Ex. 1007, 8 (teaching PACE modified RNA exhibits "enhanced nuclease resistance); Ex. 1010, 7 (teaching "cellular uptake was notably more efficient" with 2'-O-methyl thioPACE modified nucleotides). As Petitioner points out, these are the among the same benefits that Pioneer Hi-Bred suggests chemical modification of gRNA can achieve. *See* Ex. 1006, 107. As explained above, Patent Owner's arguments against Petitioner's showing for combining the references are unavailing. *See supra* § III.F.1.

Petitioner has also sufficiently shown that a POSA would reasonably expect PACE and thioPACE modifications to gRNA in a CRISPR/Cas system would be successful. As explained above, we credit the supporting testimony of Dr. Furneaux and other evidence cited in the Petition over the competing testimony of Dr. Marshall in this regard. *See supra* § III.F.1.

Patent Owner offers the same objective indicia evidence discussed in § III.F.2 above for the claims in Ground 3. As explained there, the weight of this evidence is limited because Patent Owner has not established a sufficient nexus. Considering the totality of the evidence, including objective indicia of non-obviousness, we determine that Petitioner has established, by a preponderance of the evidence, that claims 8, 11, 16, 19, and 26 would have been obvious over the combination of Pioneer Hi-Bred and Threlfall or Deleavey.[25]

H. Ground 4: Obviousness of claims 3 and 4 over Pioneer Hi-Bred in View of the Skill of a POSA

To the extent they are not anticipated, Petitioner argues claims 3 and 4 would have been obvious over Pioneer Hi-Bred in combination with the knowledge of a POSA. Pet. 74–77. More specifically, Petitioner offers evidence and argument that each of the additional limitations in these claims, i.e., a set or library comprising two or more synthetic gRNA (claim 3), and a sgRNA (claim 4), was well known in the art and would have been obvious to implement in view of Pioneer Hi-Bred's disclosure. *Id.* (citing Ex. 1003 ¶¶ 338–49).

Patent Owner advances the global arguments addressed above with respect to Petitioner's other grounds, but does not specifically dispute that the additional limitations in claims 3 and 4 would have been known to a POSA. *See* Resp. 66.

Petitioner's contentions are sufficiently supported by the record and persuasive. As explained above, Pioneer Hi-Bred anticipates claims 3 and 4 because it discloses the additional limitations in these claims and a POSA would have immediately envisioned embodiments meeting the same. To the extent one might argue otherwise, Petitioner has shown that claims 3 and 4 would have been obvious because the additional elements they recite were

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[25] Even if Patent Owner had established a sufficient nexus between these claims and its industry praise and copying evidence such that it was entitled to more weight, we would reach the same conclusion given the relative strength of Petitioner's showing.

well known to a POSA and taught in Pioneer Hi-Bred. In this regard, we credit Petitioner's contentions and the supporting the testimony of Dr. Furneaux (i.e., Ex. 1003 ¶¶ 338–49). Thus, considering the totality of the evidence, including objective indicia of non-obviousness, we determine that Petitioner has established, by a preponderance of the evidence, that claims 3 and 4 would have been obvious over Pioneer Hi-Bred in view of the skill of a POSA.

I. Ground 5: Obviousness of claims 5, 8–13, 20, 21, and 24–28 over Pioneer Hi-Bred in View of the Skill of a POSA

Petitioner further asserts that claims 5, 8–13, 20, 21, and 24–28 would have been obvious over Pioneer Hi-Bred in view of the skill of a POSA. Pet. 77–85.

Having already determined that these claims are anticipated for the reasons in Ground 1 and obvious for the reasons in Ground 2, we need not decide Petitioner's additional challenge to these claims in Ground 5. *See SAS Inst. Inc. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding that a petitioner "is entitled to a final written decision addressing all of the claims it has challenged"); *Boston Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App'x 984, 990 (Fed. Cir. 2020) (nonprecedential) ("We agree that the Board need not address [alternative grounds] that are not necessary to the resolution of the proceeding.").

J. Ground 6: Obviousness of claims 14 and 29 over Pioneer Hi-Bred in View of the Skill of a POSA

To the extent they are not anticipated, Petitioner argues claims 14 and 19 would have been obvious over Pioneer Hi-Bred in combination with the knowledge of a POSA. Pet. 85–86. More specifically, Petitioner offers evidence and argument that the additional limitations in these claims, i.e., a fluorophore at a 5' end, was well known in the art and would have been obvious to implement in view of Pioneer Hi-Bred's disclosure. *Id.* (citing Ex. 1003 ¶¶ 391–95).

Patent Owner advances the global arguments addressed above with respect to Petitioner's other grounds, but does not specifically dispute that the additional limitations in claims 14 and 29 would have been known to a POSA. *See* Resp. 66–67.

Petitioner's contentions are sufficiently supported by the record and persuasive. As explained above, Pioneer Hi-Bred anticipates claims 14 and 29 because it discloses the addition of a fluorophore to the VT domain and a POSA would have understood that the VT domain may be at either the 3' or 5' end. *Supra* § III.E.vi. Thus, a POSA would have immediately envisioned embodiments meeting the same. To the extent one might argue otherwise, Petitioner has shown that claims 14 and 29 would have been obvious because the additional elements they recite were well known to a POSA and taught in Pioneer Hi-Bred. In this regard, we credit Petitioner's contentions and the supporting the testimony of Dr. Furneaux (i.e., Ex. 1003 ¶¶ 391–95). Thus, considering the totality of the evidence, including objective indicia of non-obviousness, we determine that Petitioner has established, by a preponderance of the evidence, that claims 14 and 29 would have been obvious over Pioneer Hi-Bred in view of the skill of a POSA.

IV. MOTIONS TO SEAL

The parties jointly request that Exhibits 1053–1058 and portions of Petitioner's Reply discussing those exhibits be sealed. Paper 29 ("Joint

Motion”). These exhibits “are excerpts of the deposition of Dr. Daniel Ryan, a named inventor on the . . . ’034 [patent], along with certain exhibits to his deposition transcript.” *Id.* at 1–2. “Patent Owner avers that the documents from the Ryan deposition provide information on proprietary research, as well as confidential information about Patent Owner’s business practices.” *Id*. at 3. The parties also represent that the district court in a related proceeding “explicitly ruled that the Ryan transcript and exhibits that are the subject of this Motion to Seal have been properly designated as confidential.” *Id.* (quotations omitted).

Along with their motion, the parties submit a proposed protective order. *Id.* at App. A. This order follows the guidelines in Appendix B of the Trial Practice Guide,[26] but modifies the highly confidential designation “by eliminating access by ‘persons who are named parties to the proceeding,’ ‘party representatives,’ and ‘in-house counsel.’” *Id.* at 4–5. The parties represent that this modification aligns the designations in this proceeding with those that have been entered in the related district court proceeding and therefore “will aid in the efficient administration of justice, and will be more straightforward and expedient than preparing a ‘two level’ protective order for” the parties’ IPR proceedings. *Id.* at 5.

A party may move to seal confidential information including, *e.g.*, confidential research, development, or commercial information. TPG 19; 37 C.F.R. § 42.54. It is the movant’s burden to show good cause for sealing such information, and we balance the party’s asserted need for confidentiality with the strong public interest in open proceedings. *Argentum*

[26] Consolidated Trial Practice Guide (Nov. 2019), 64, available at https://www.uspto.gov/sites/default/files/documents/tpgnov.pdf (“TPG”).

Pharms. LLC v. Alcon Research, Ltd., IPR2017-01053, Paper 27 at 4 (PTAB Jan. 19, 2018) (informative).

The parties provide a sufficient explanation and have shown good cause for sealing exhibits 1053–1058 and the related portions of Petitioner's Reply. Moreover, Petitioner provides a public version of its Reply (Paper 31) with redactions limited to a few paragraphs discussing these exhibits so the record may remain clear and reasonably open. We also determine that the proposed protective order with the agreed modification limiting access to outside counsel and the parties' experts is appropriate under these circumstances.

Patent Owner additionally moves to seal a paragraph in its Sur-reply, "which discusses documents as to which sealing was jointly requested" in the Joint Motion. Paper 34. Petitioner provides a public version of its Sur-reply (Paper 35) with redactions limited to a single paragraph. We find that Patent Owner has shown good cause for granting its motion.

Accordingly, Exhibits 1053–1058, Petitioner Reply (Paper 30), and Patent Owner's Sur-reply (Paper 33) are sealed, and the protective order, Appendix A to Paper 29, is entered.

V. CONCLUSION[27]

Petitioner has shown, by a preponderance of the evidence, that claims 1–33 of the '034 patent are unpatentable.

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[27] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this Decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or*

Claims	35 U.S.C. §	Reference(s)/ Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1–5, 8–21, 24–33	102	Pioneer Hi-Bred	1–5, 8–21, 24–33	
5, 8–13, 18, 20, 21, 24–28, 32	103	Pioneer Hi-Bred, Krutzfeldgt, Deleavey, Soutschek, Yoo	5, 8–13, 18, 20, 21, 24–28, 32	
6, 7, 22, 23	103	Pioneer Hi-Bred, Threlfall, Deleavey	6, 7, 22, 23	
3, 4	103	Pioneer Hi-Bred	3, 4	
5, 8–13, 20, 21, 24–28[28]	103	Pioneer Hi-Bred		
14, 29	103	Pioneer Hi-Bred	14, 29	
Overall Outcome			1–33	

Reexamination During a Pending AIA Trial Proceeding. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. §§ 42.8(a)(3), 42.8(b)(2).

[28] As explained above, we do not reach this ground.

VI. ORDER

Accordingly, it is:

ORDERED that Petitioner has shown that claims 1–33 of U.S. Patent 10,900,034 B2 are unpatentable;

FURTHER ORDERED that the Joint Motion to Seal (Paper 29) and Patent Owner's Motion to Seal Portions of the Sur-reply (Paper 33) are *granted*; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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